

Assignment 1: Quality Assessment, Quality Trimming, and Taxonomic Classification of 16S rRNA Sequencing Data

Group Number	3	
Pipeline / Tools	FastQC, Trimmomatic (0.41), SPINGO	
Dataset	12 paired-end 16S rRNA samples (24 FASTQ files)	
Total Marks	20 (Q1: 5, Q2: 5, Q3: 5, Q4: 5)	
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Date	30.09.2026	
Subject	COMPUTATIONAL METAGENOMICS	

Table of Contents

1. Question 1 — Initial Quality Check with FastQC
2. Question 2 — Quality Trimming with Trimmomatic
3. Question 3 — Taxonomic Classification with SPINGO
4. Question 4 — Comparison of Genus- and Species-Level Profiles
5. Group Contribution Summary

1. In this section you will perform an initial quality check for the sequences using FastQC pipeline (this step can be done using windows system as well). Show the plots having the quality scores and add small snapshots for each sample. (5 marks)

Question 1: Initial Quality Check (FastQC)

The raw 16S rRNA paired-end reads for all 12 samples (24 FASTQ files: SRR14919235 to SRR14919246, each with a forward (_1.fastq.gz) and reverse (_2.fastq.gz) were assessed for sequencing quality using FastQC prior to any trimming. FastQC reports were generated individually for every file and then merged into a single overview using MultiQC, which allows quality metrics to be compared across all samples simultaneously.

Method: A dedicated Conda environment (assignment1) was created with FastQC, Trimmomatic and MultiQC installed from the bioconda channel. FastQC was executed on all 24 raw FASTQ files using 4 threads, and the 24 resulting HTML/ZIP reports were combined with MultiQC into a single interactive summary report. The exact commands used are documented in the accompanying README.md file.

Sequence Quality Histograms

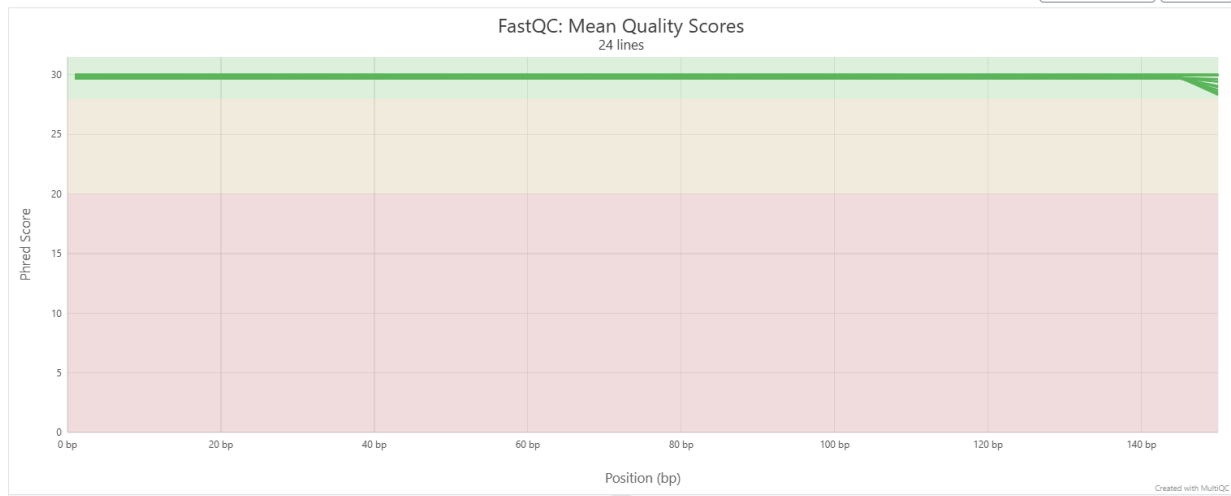
24

Help

The mean quality value across each base position in the read.

+ Summarize plot

Export...



Per Sequence Quality Scores

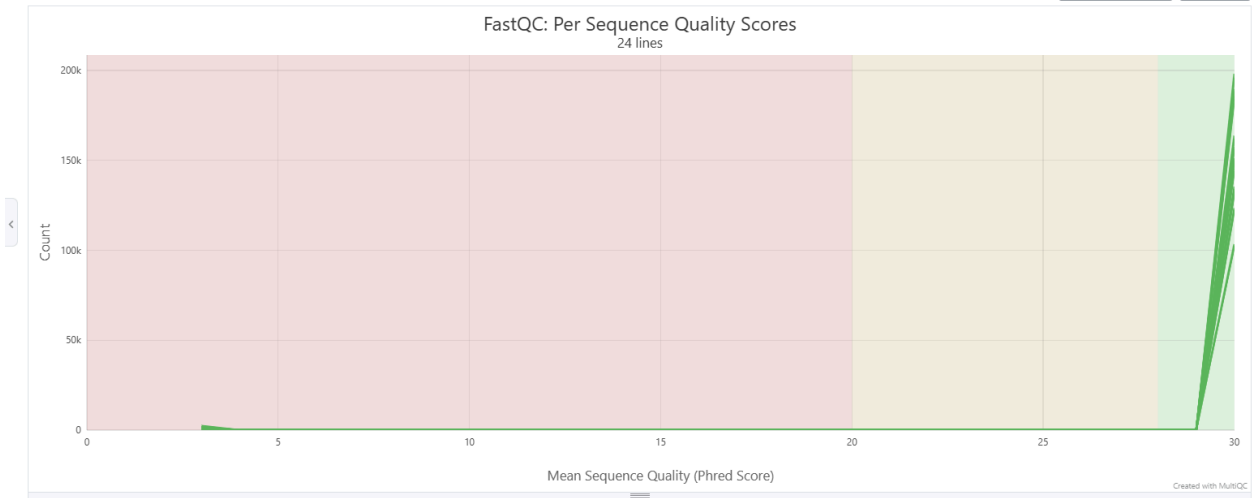
24

Help

The number of reads with average quality scores. Shows if a subset of reads has poor quality.

Summarize plot

Export...



Per Sequence GC Content

24

Help

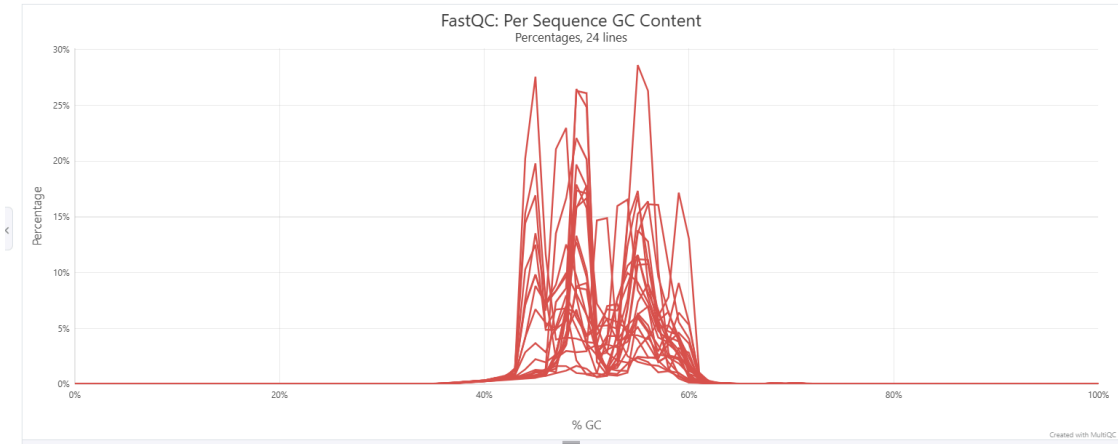
The average GC content of reads. Normal random library typically have a roughly normal distribution of GC content.

Percentages

Counts

Summarize plot

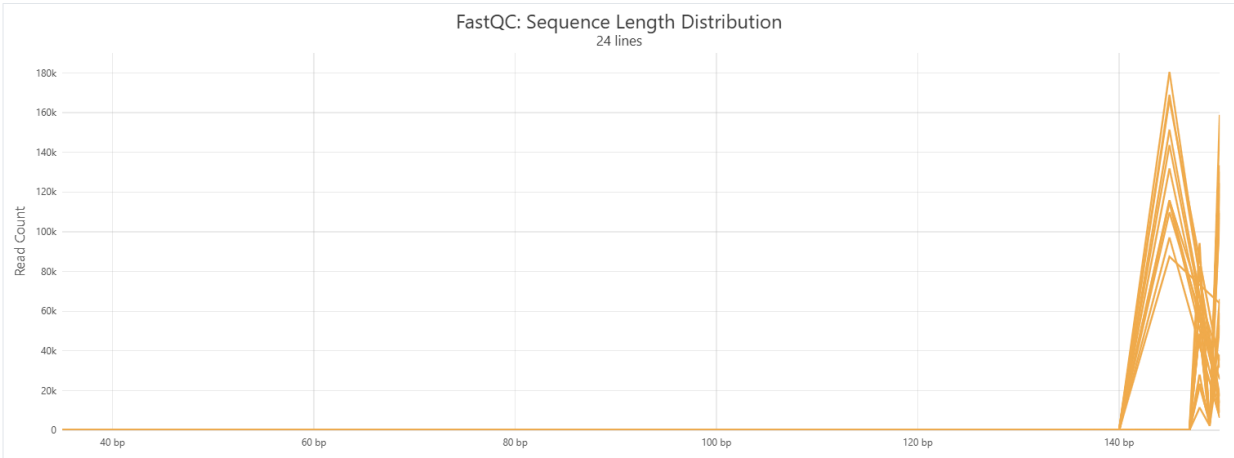
Export...



Sequence Length Distribution 24

The distribution of fragment sizes (read lengths) found. See the [FastQC help](#)

Summarize plot Export...

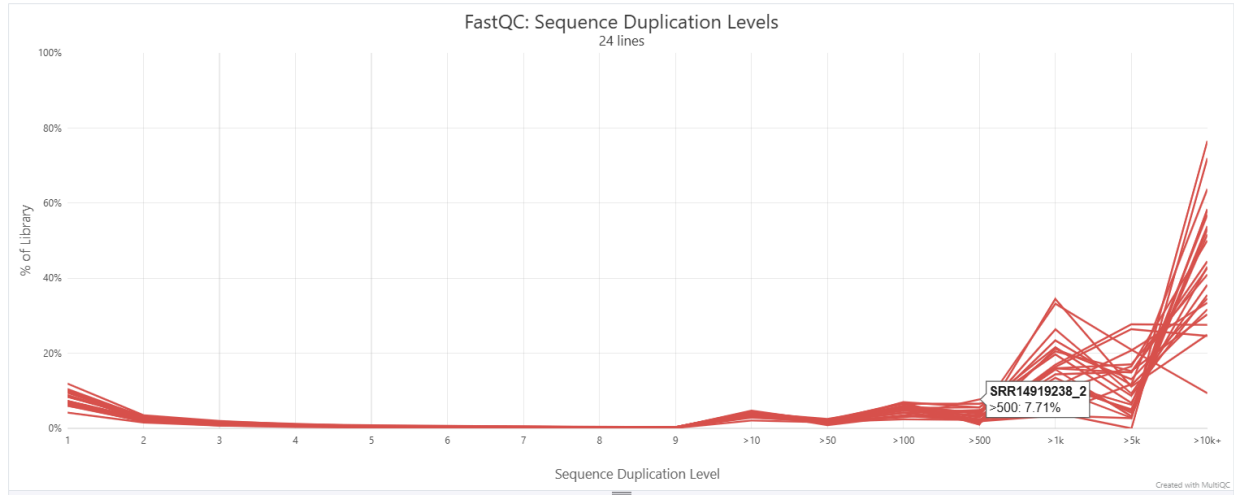


Sequence Duplication Levels 24

The relative level of duplication found for every sequence.

Help

Summarize plot Export...

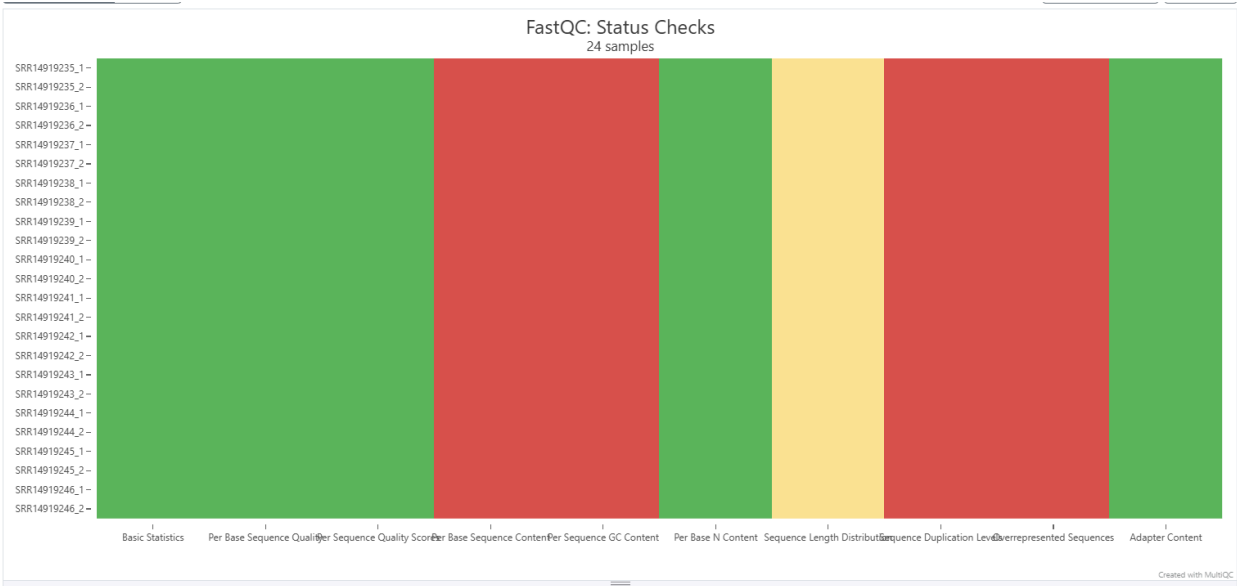
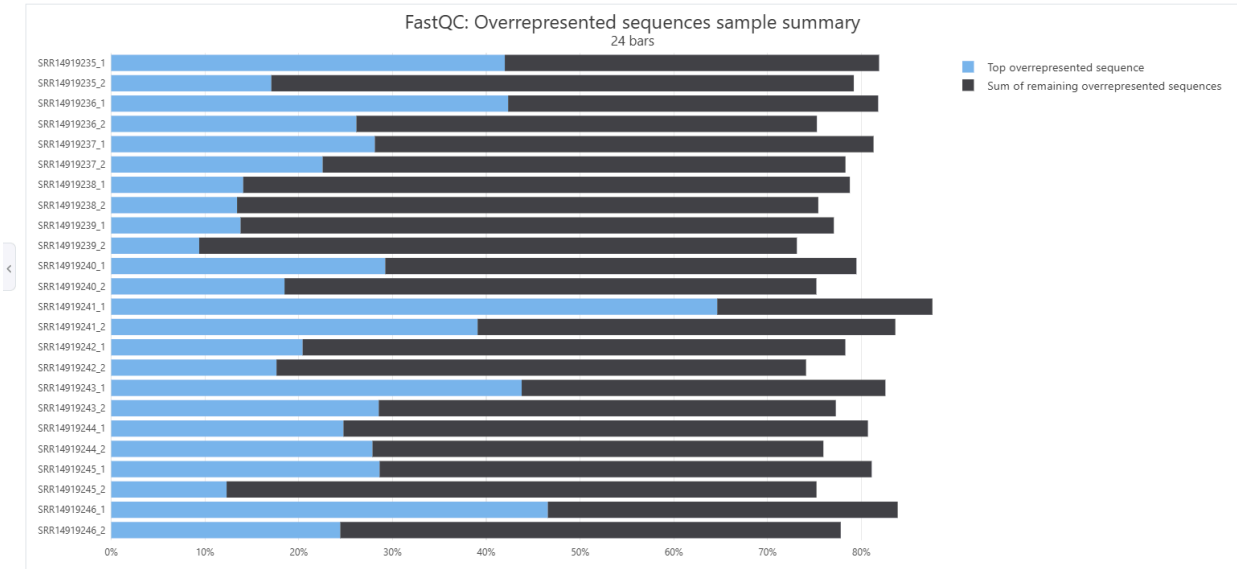


Overrepresented sequences by sample

Help

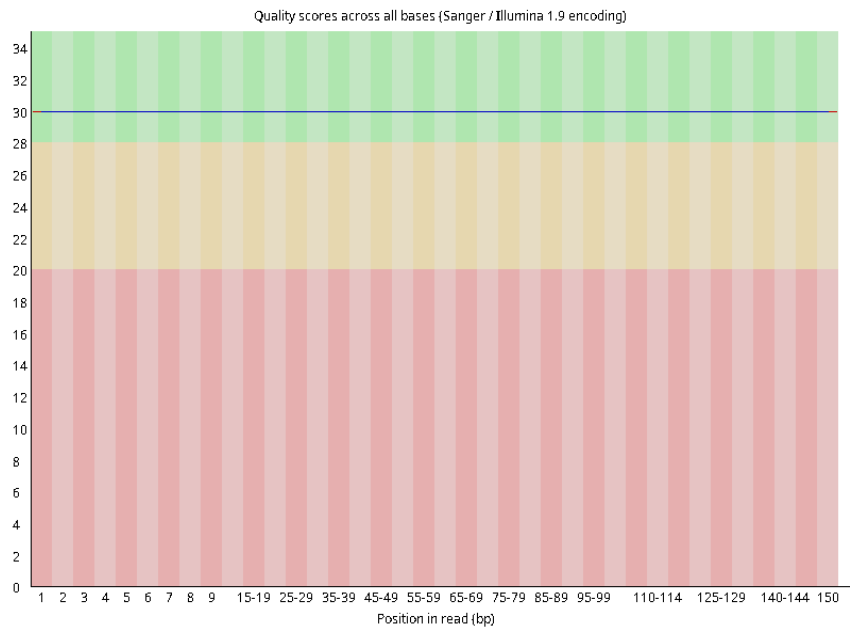
The total amount of overrepresented sequences found in each library.

Summarize plot Export...

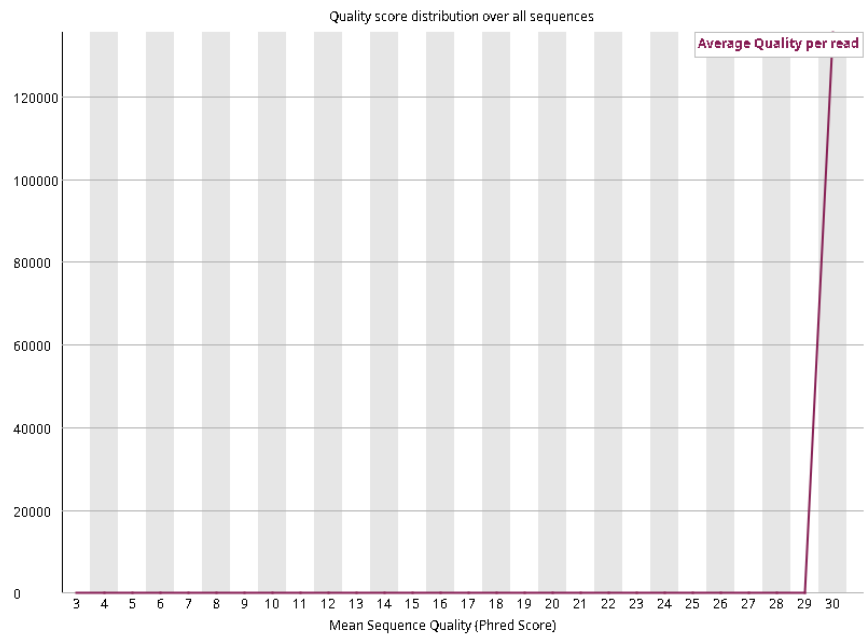


Each Sample
SRR14919235
Forward

✔ Per base sequence quality

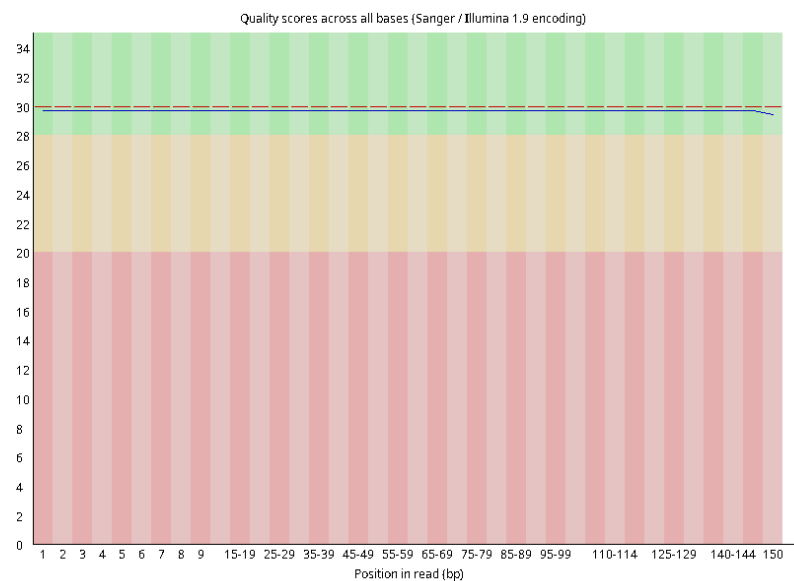


✔ Per sequence quality scores

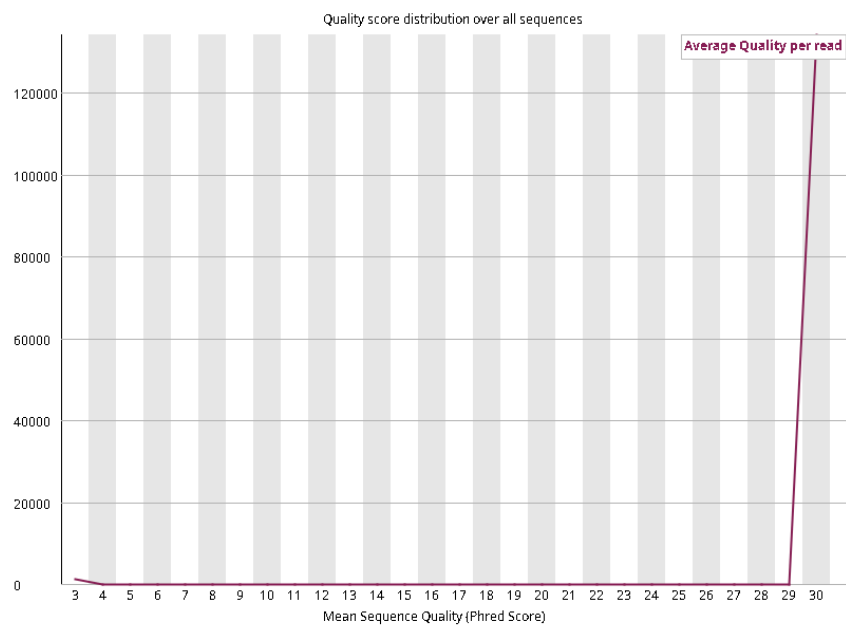


Reverse

✔ Per base sequence quality



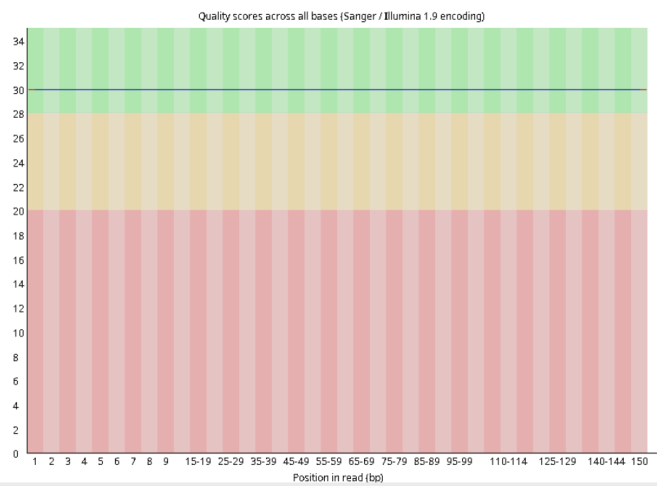
✔ Per sequence quality scores



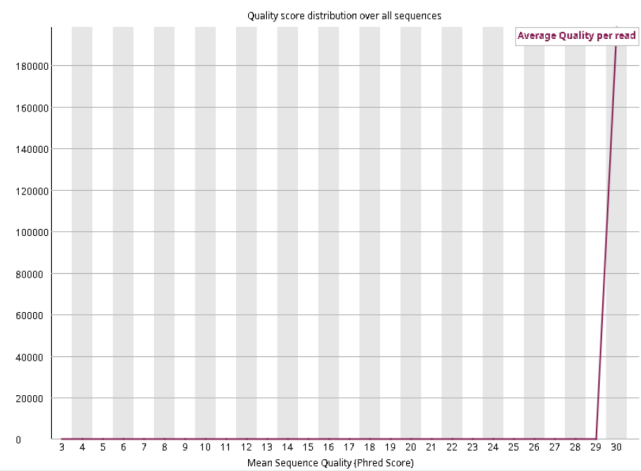
SRR14919236

Forward:

✔ Per base sequence quality

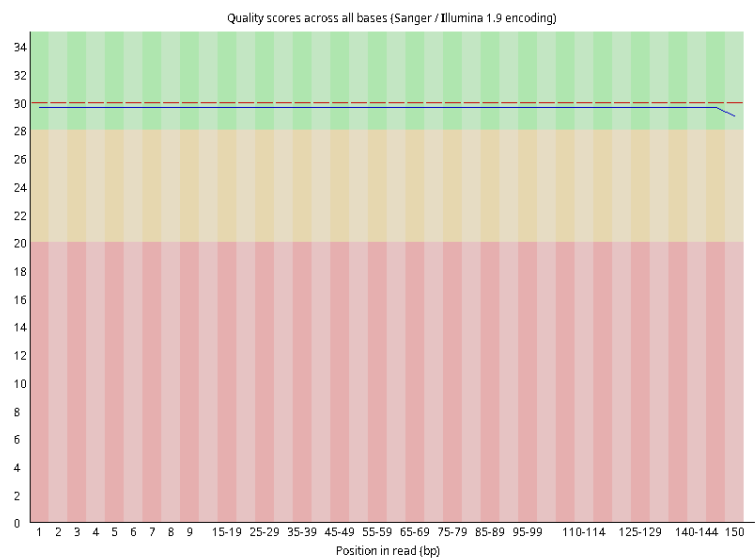


✔ Per sequence quality scores

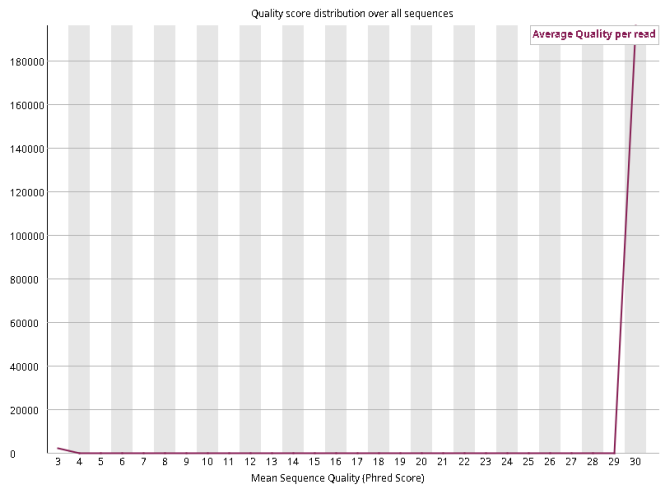


Reverse

✔ Per base sequence quality



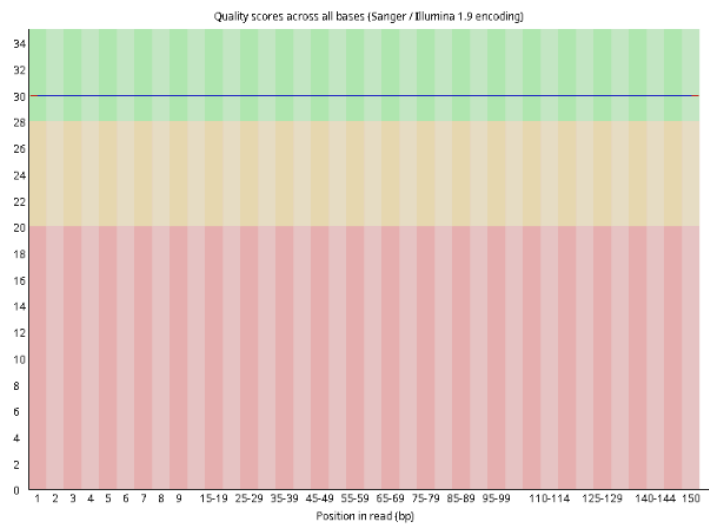
✔ Per sequence quality scores



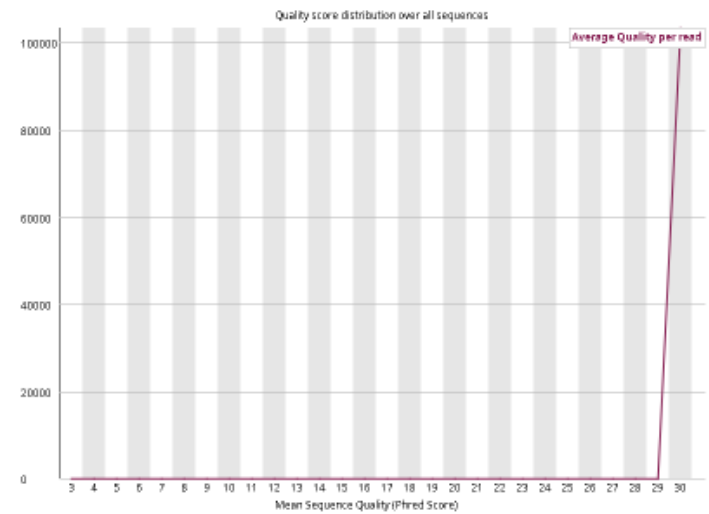
SRR14919237

Forward:

✔ Per base sequence quality

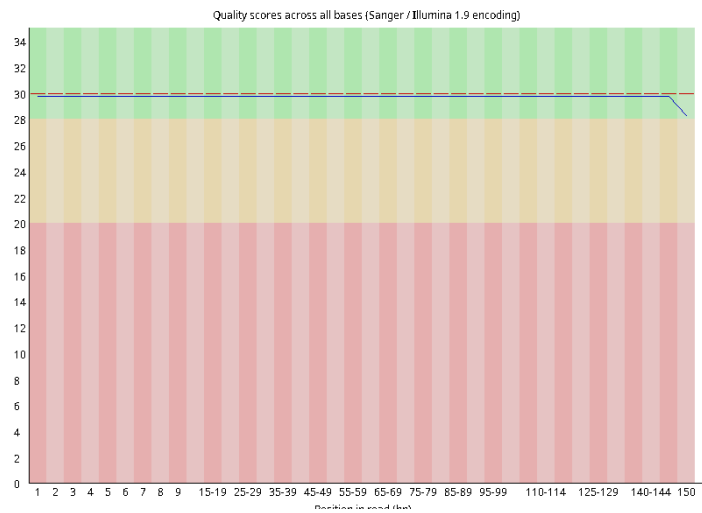


✔ Per sequence quality scores

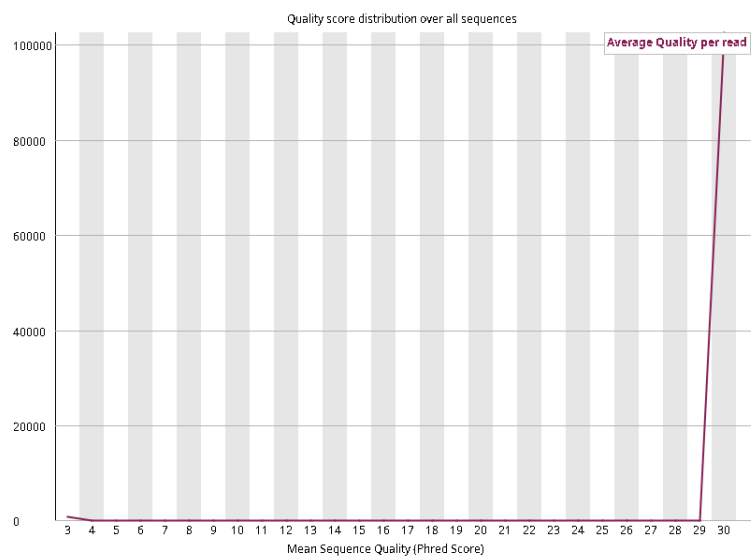


Reverse:

✔ Per base sequence quality



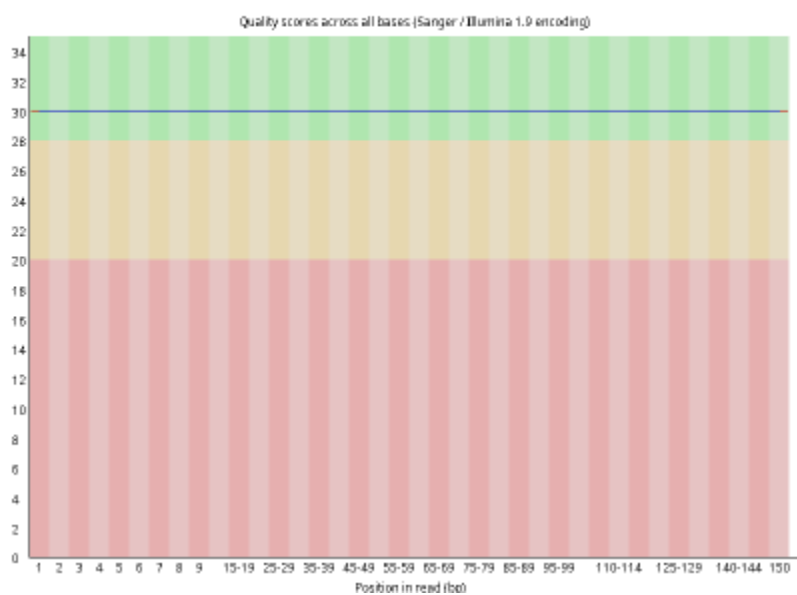
✔ Per sequence quality scores



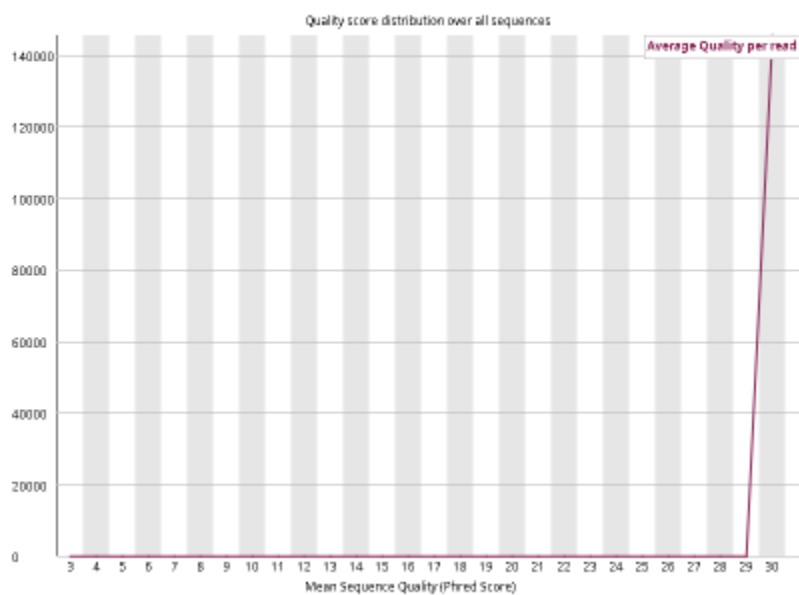
SRR14919238

Forward:

✓ Per base sequence quality

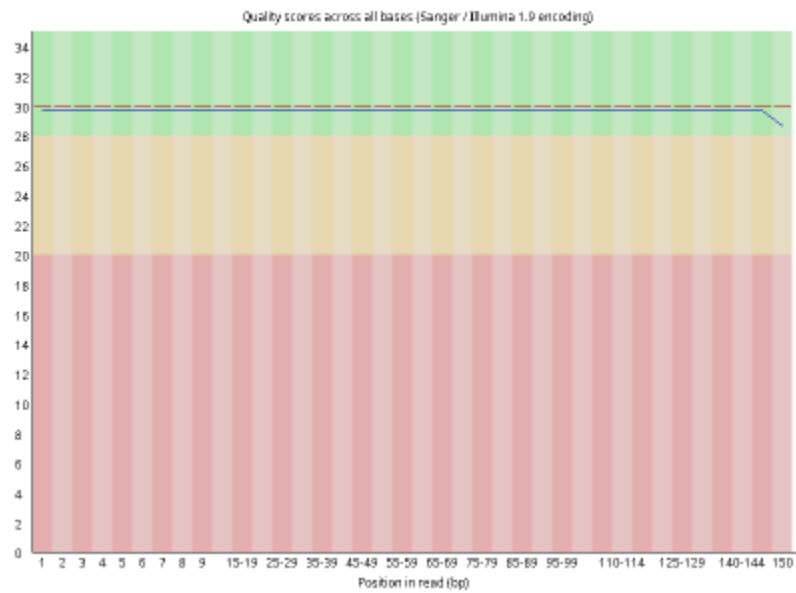


✓ Per sequence quality scores

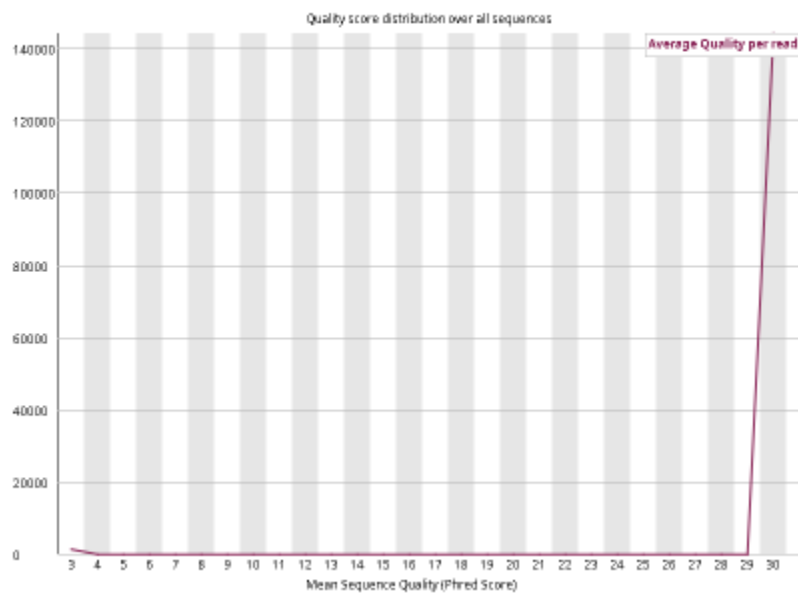


Reverse:

✓ Per base sequence quality



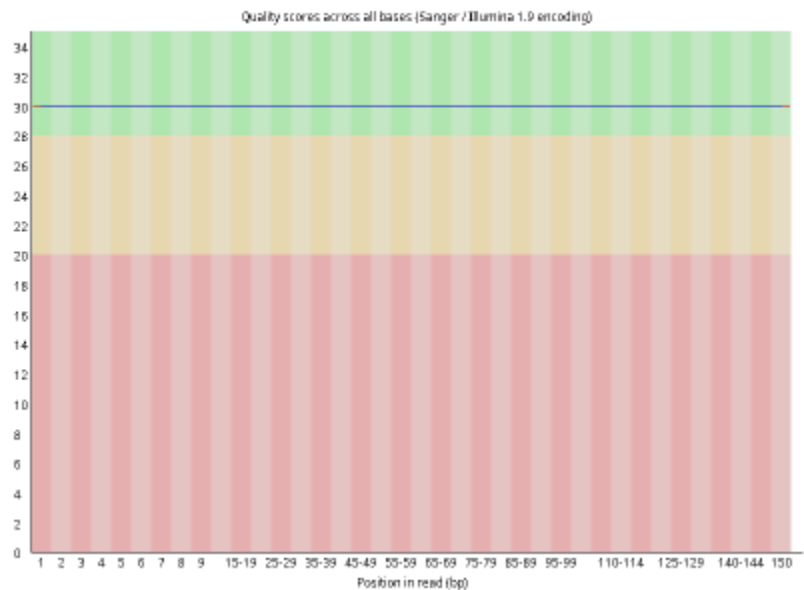
✓ Per sequence quality scores



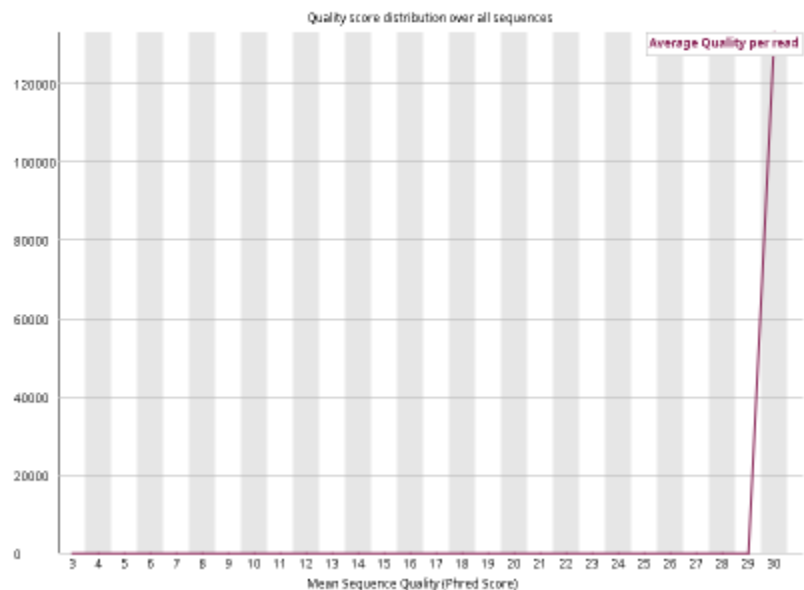
SRR14919239

Forward:

✔ Per base sequence quality

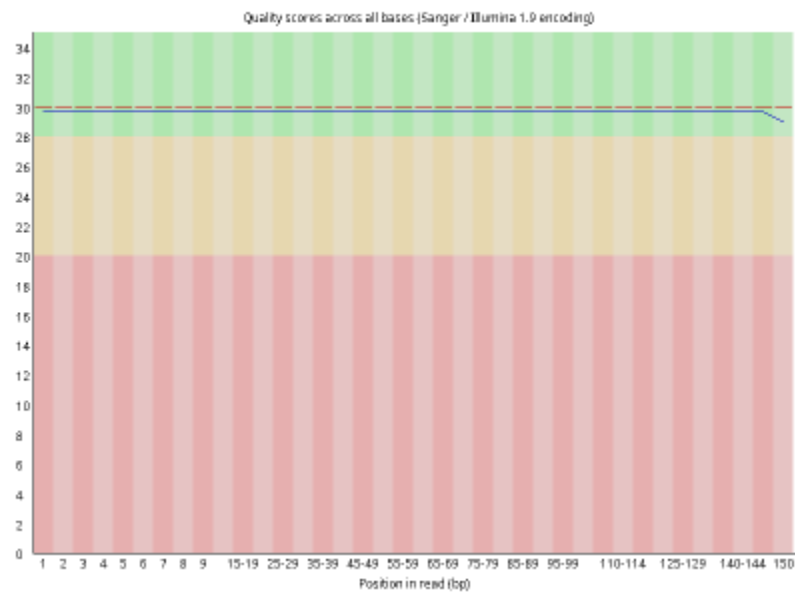


✔ Per sequence quality scores

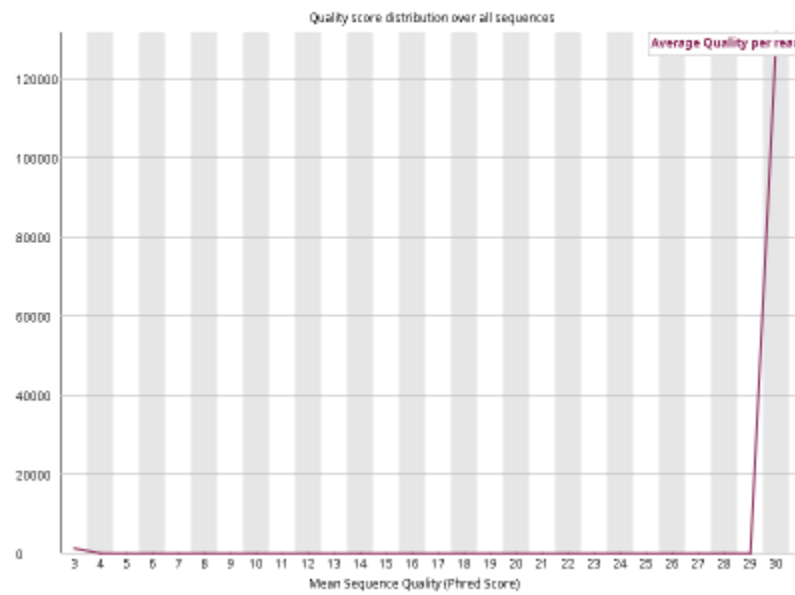


Reverse:

✔ Per base sequence quality



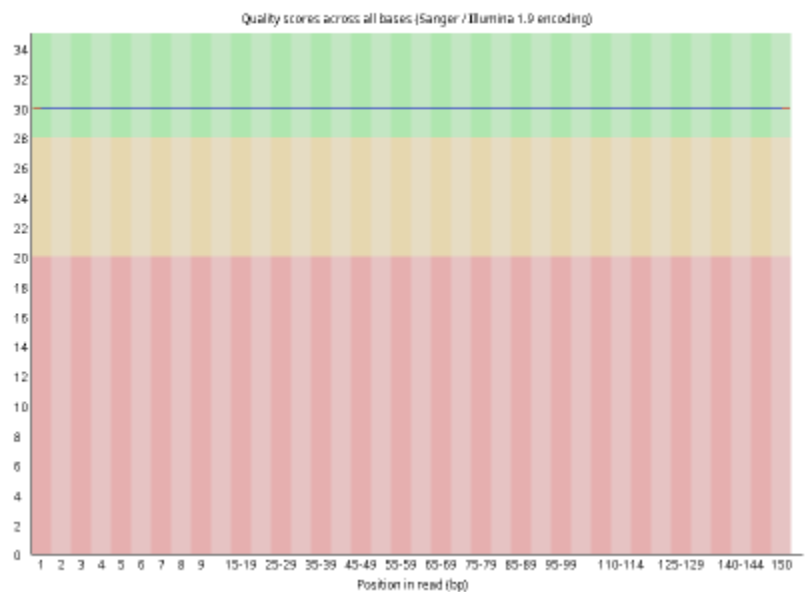
✔ Per sequence quality scores



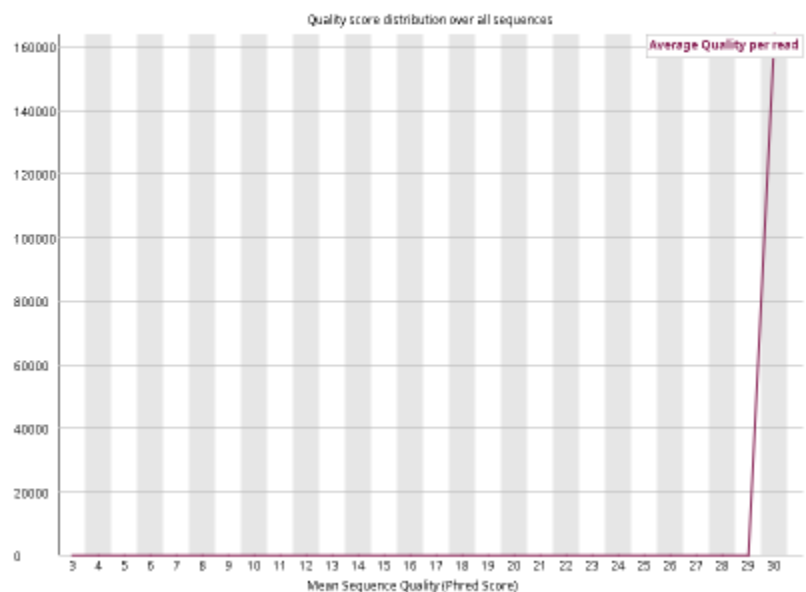
SRR14919240

Forward:

✔ Per base sequence quality

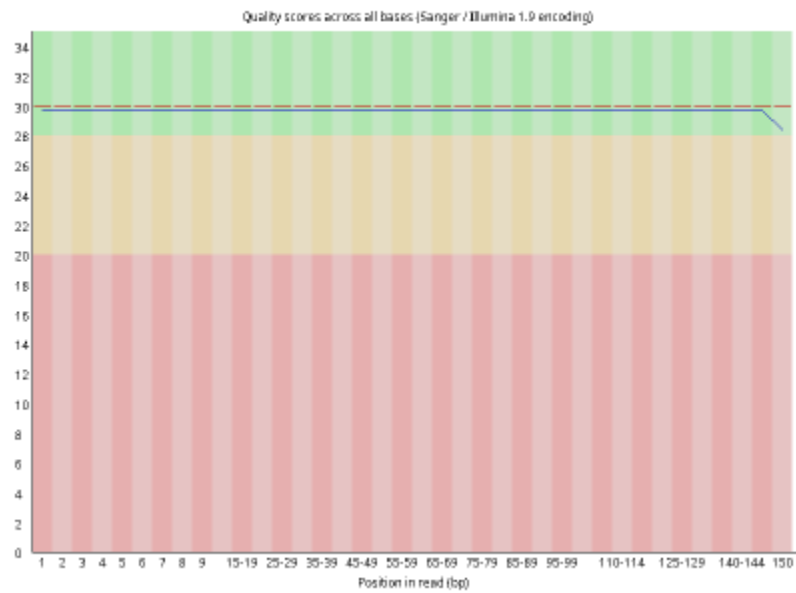


✔ Per sequence quality scores



Reverse:

✔ Per base sequence quality



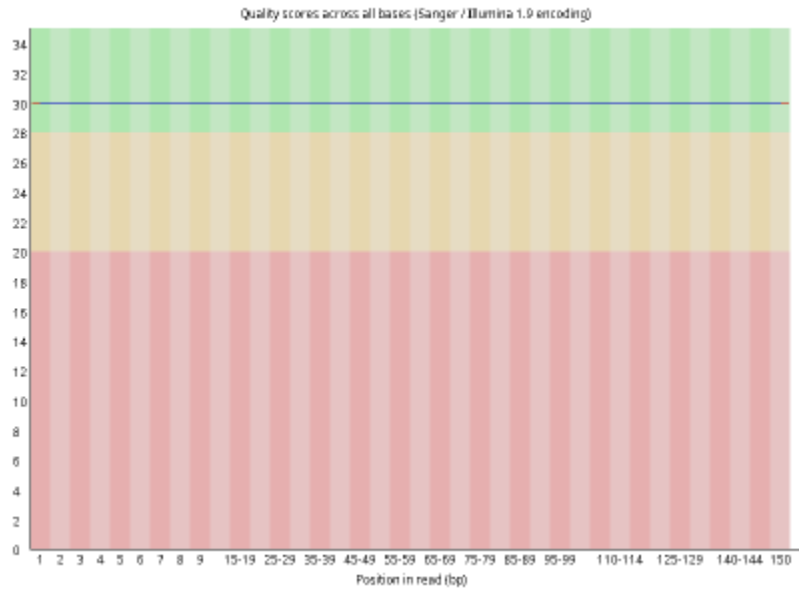
✔ Per sequence quality scores



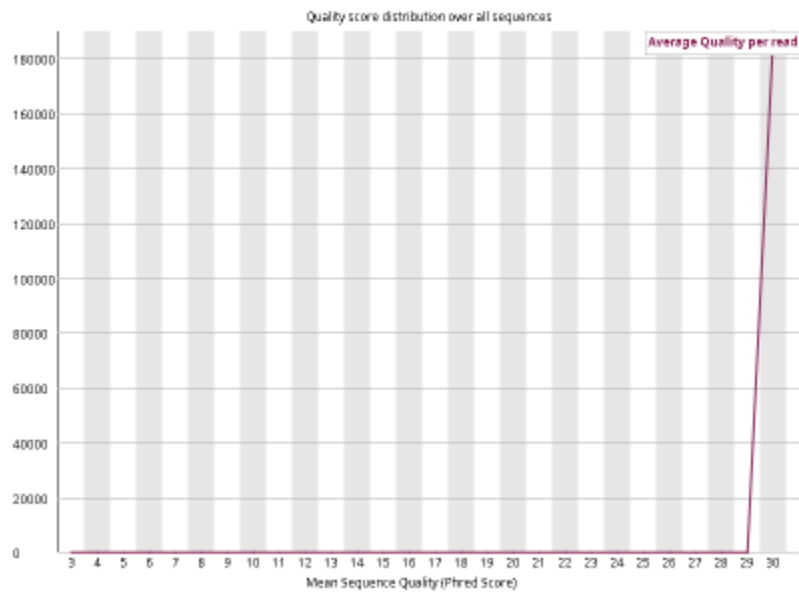
SRR14919241

Forward:

✔ Per base sequence quality

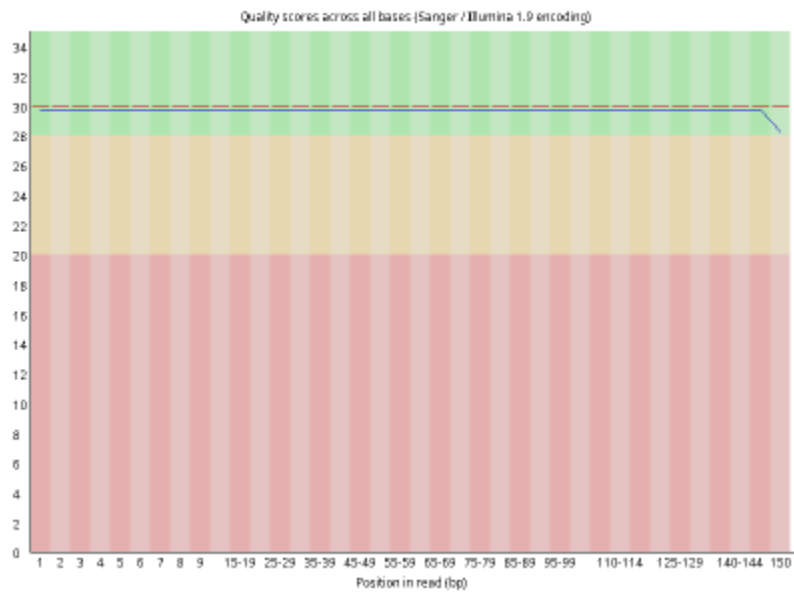


✔ Per sequence quality scores



Reverse:

✔ Per base sequence quality



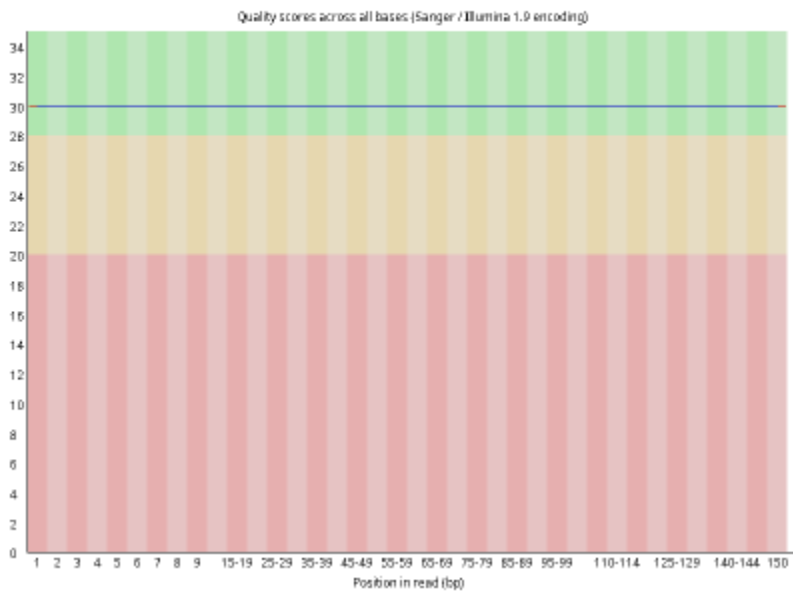
✔ Per sequence quality scores



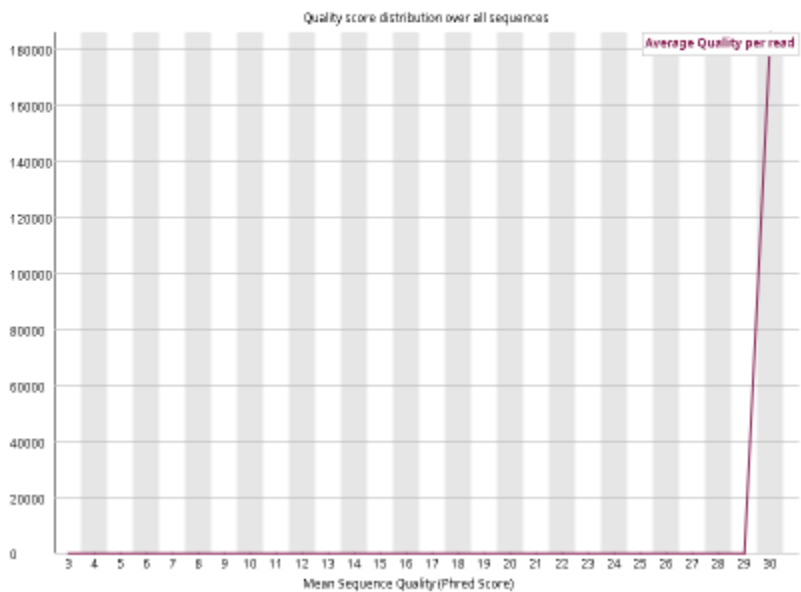
SRR14919242

Forward:

✔ Per base sequence quality

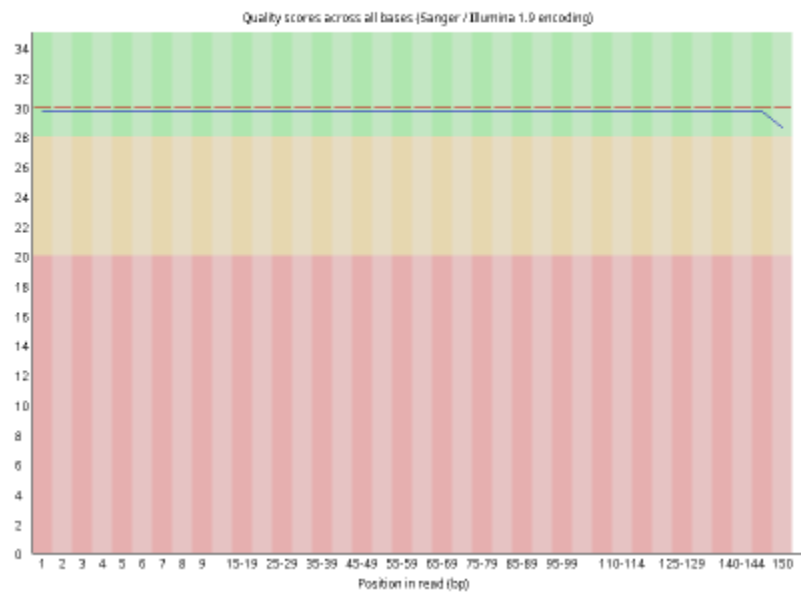


✔ Per sequence quality scores

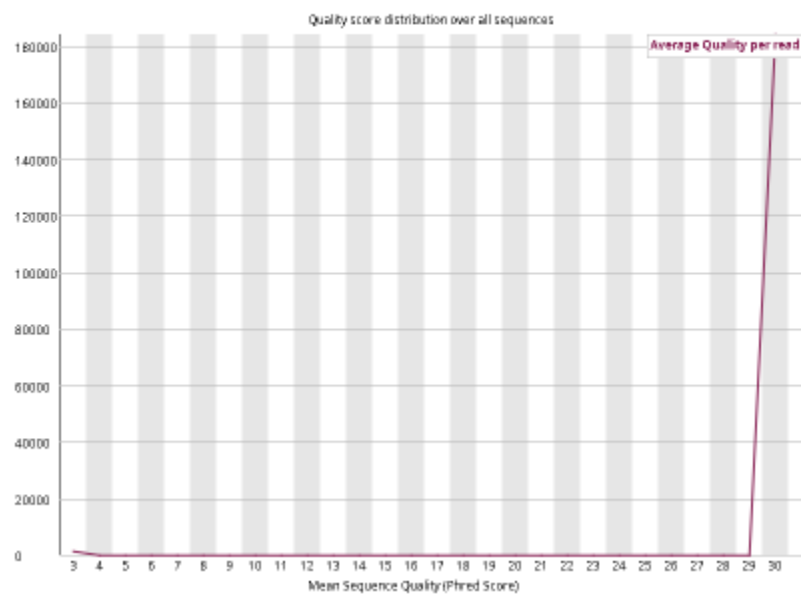


Reverse:

✓ Per base sequence quality



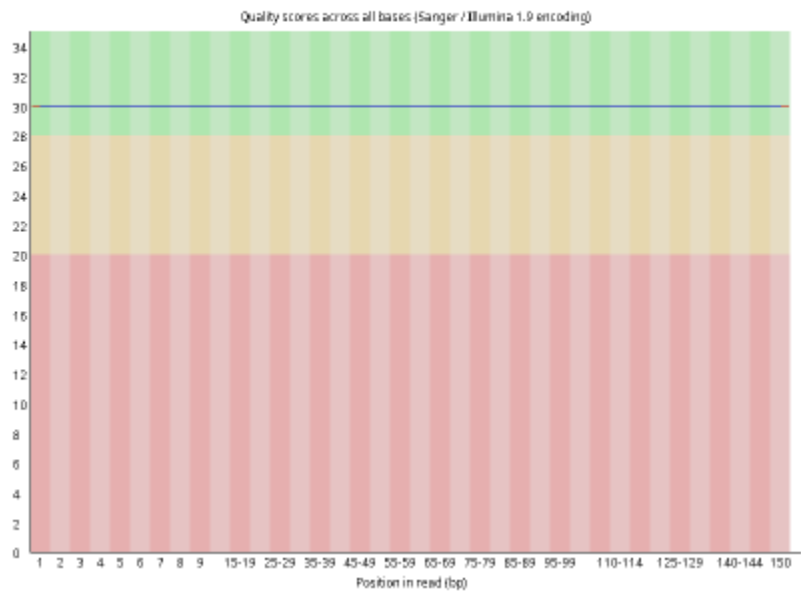
✓ Per sequence quality scores



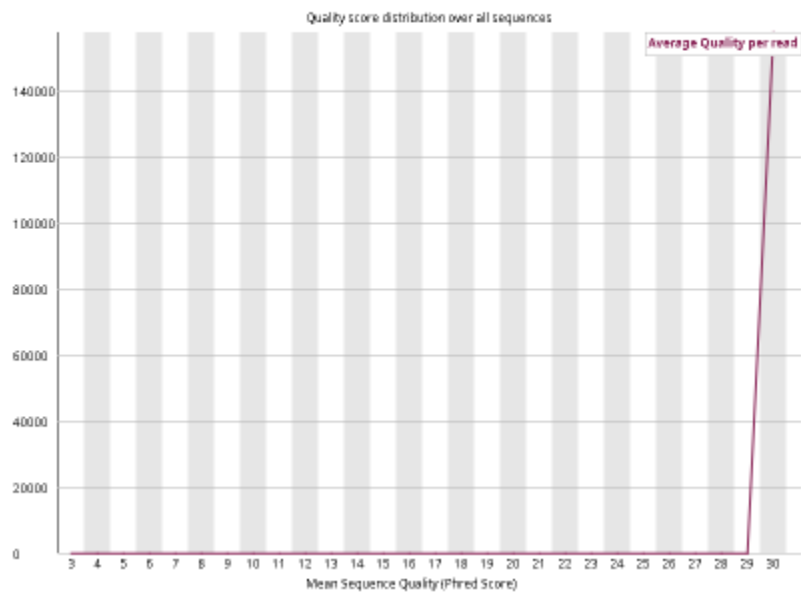
SRR14919243

Forward:

✔ Per base sequence quality

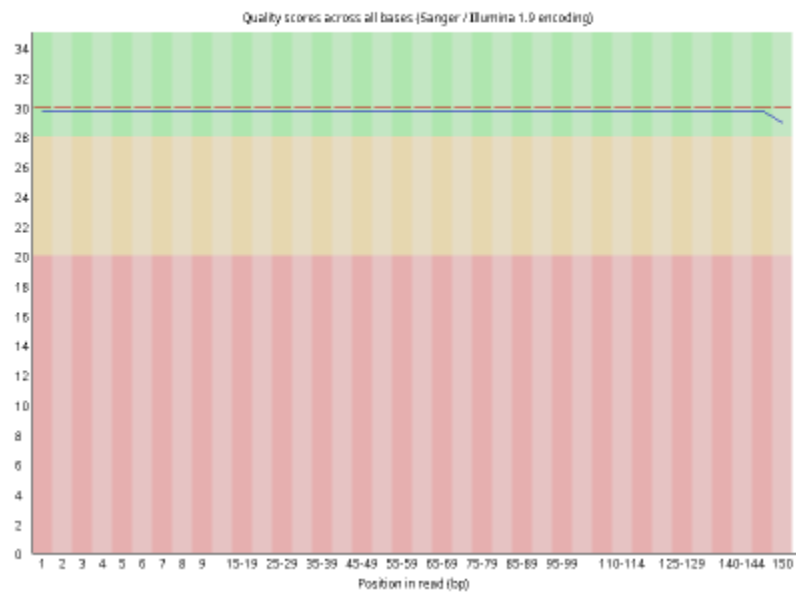


✔ Per sequence quality scores

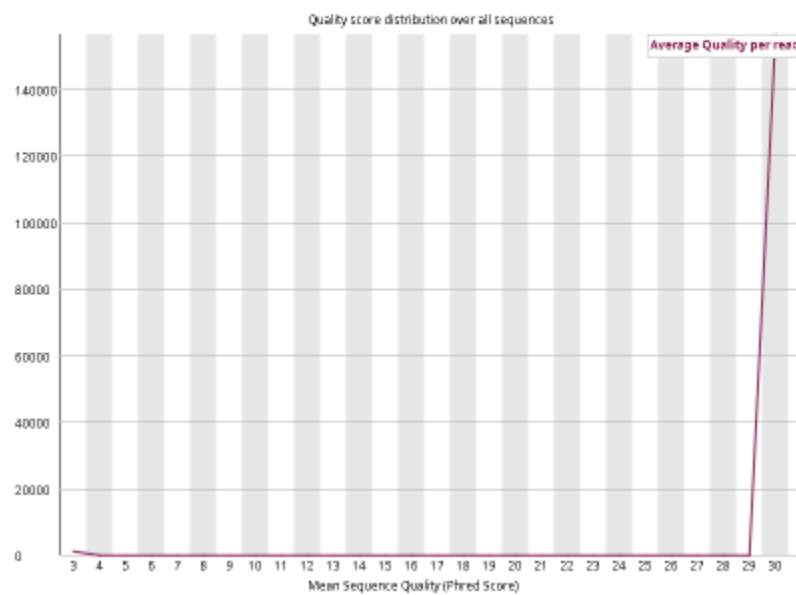


Reverse:

✓ Per base sequence quality



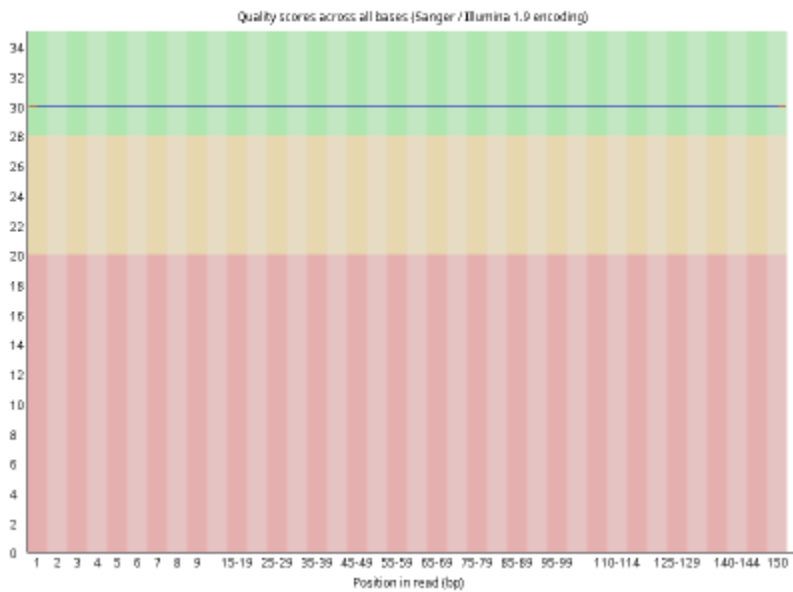
✓ Per sequence quality scores



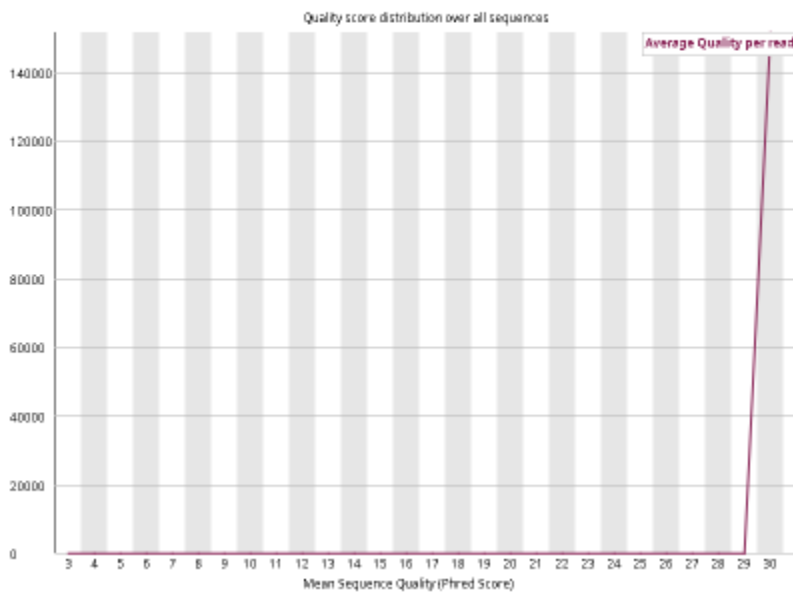
SRR14919244

Forward:

✔ Per base sequence quality

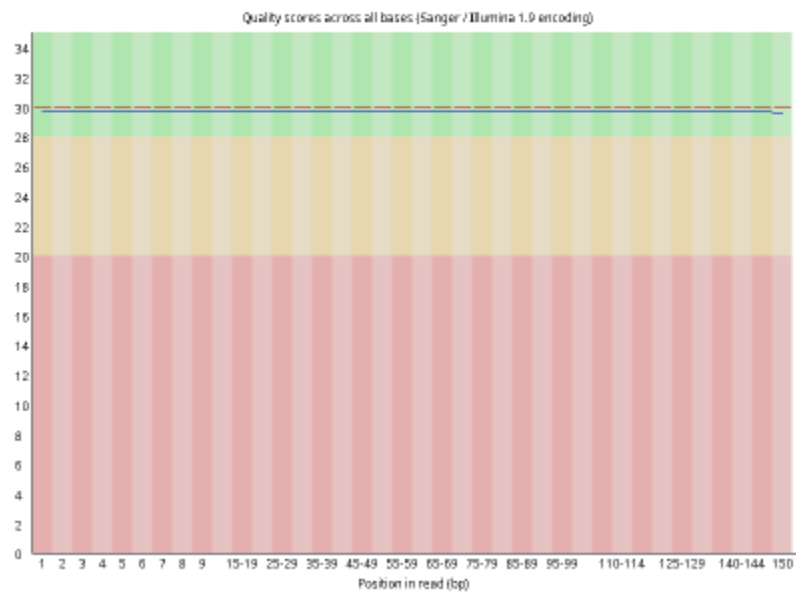


✔ Per sequence quality scores

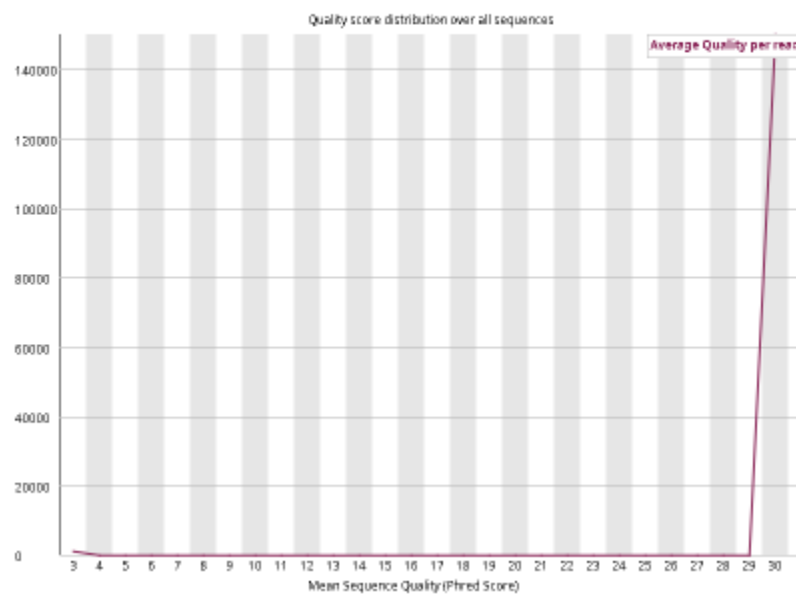


Reverse:

✔ Per base sequence quality



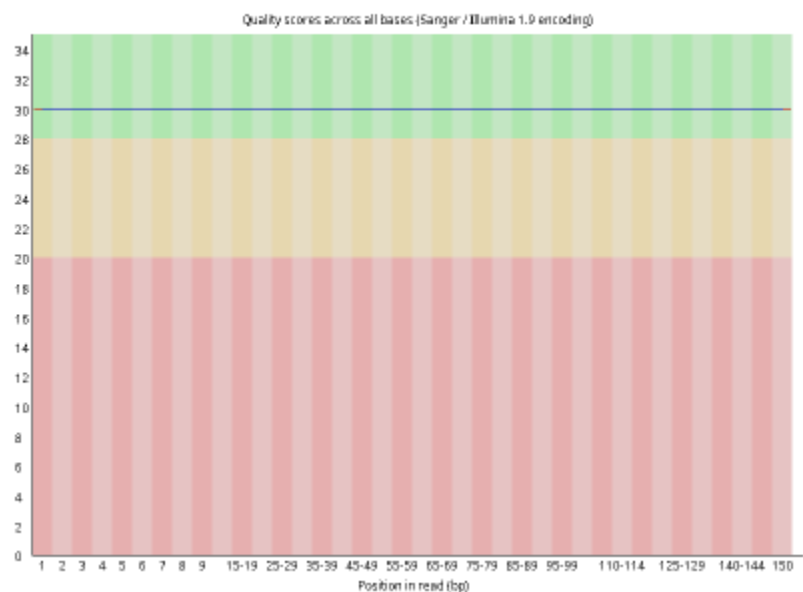
✔ Per sequence quality scores



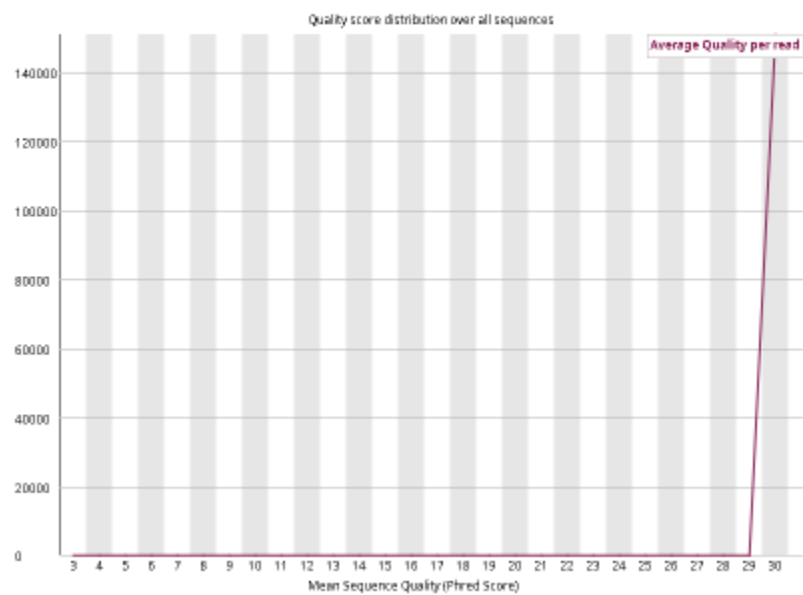
SRR14919245

Forward:

✔ Per base sequence quality

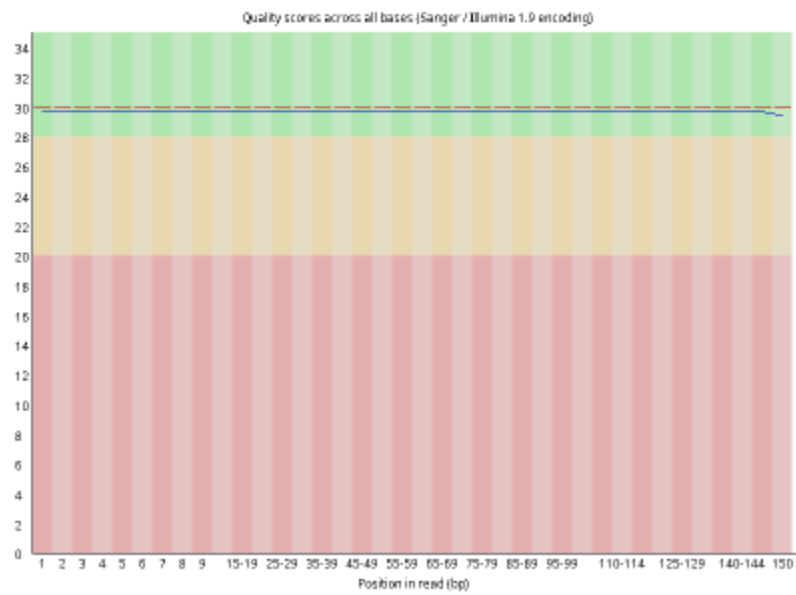


✔ Per sequence quality scores

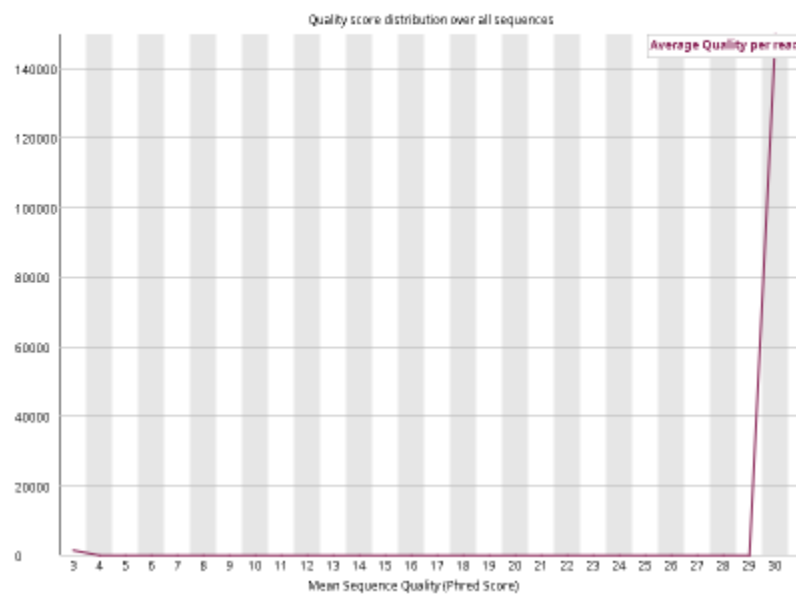


Reverse:

✓ Per base sequence quality



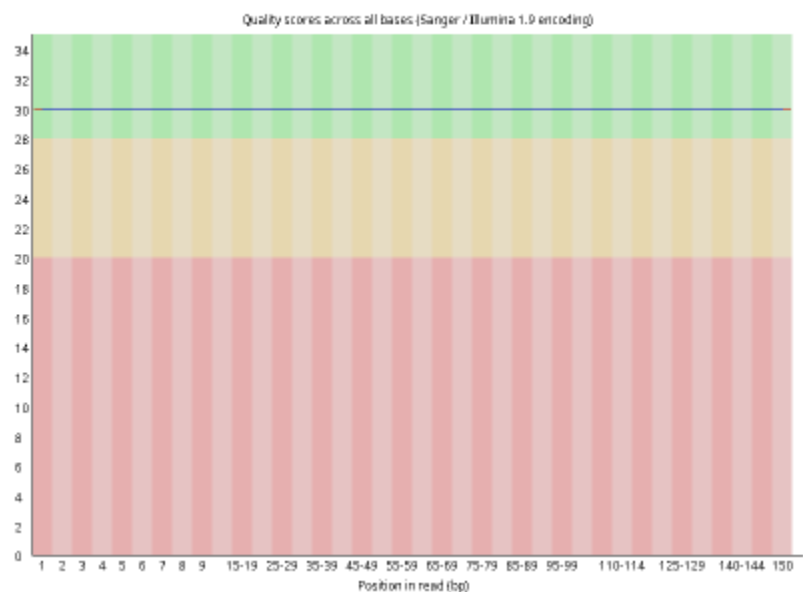
✓ Per sequence quality scores



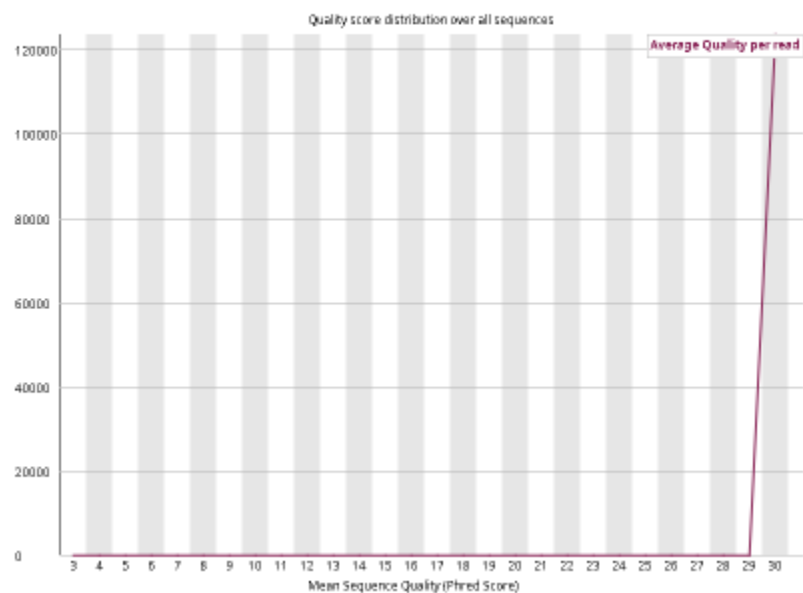
SRR14919246

Forward:

✓ Per base sequence quality

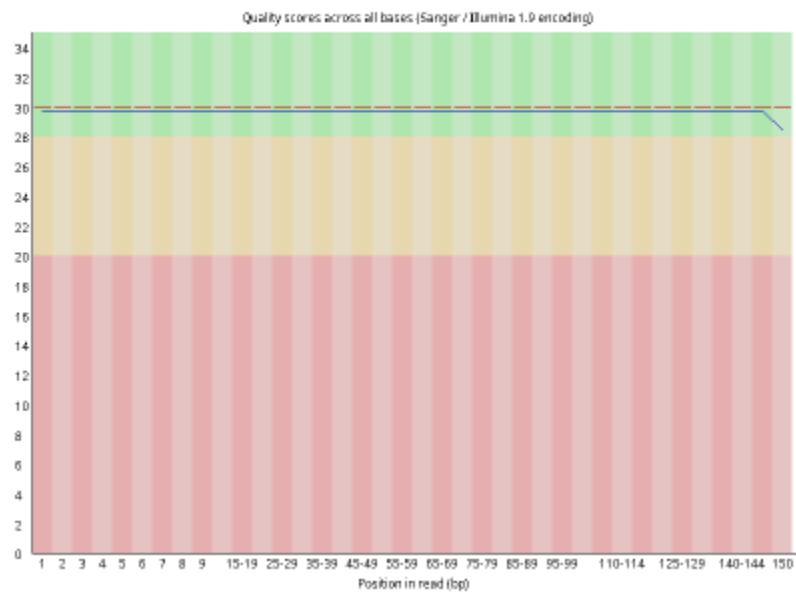


✓ Per sequence quality scores

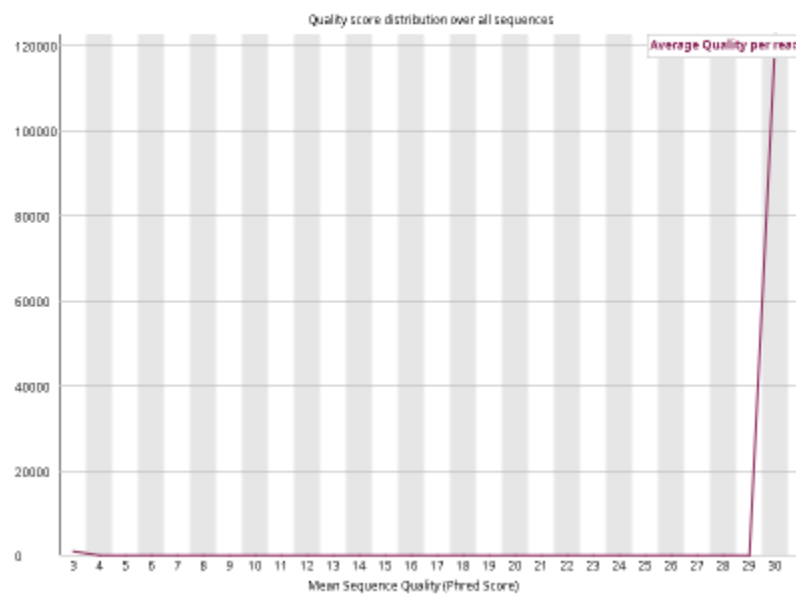


Reverse:

✔ Per base sequence quality



✔ Per sequence quality scores



2. Run trimmomatic to perform a quality trimming of the reads in the datasets. Perform the trimming with the following parameters: SLIDINGWINDOW:5:27 MINLEN:100 AVGQUAL:27. Infer the results (5 marks)

Question 2: Quality Trimming with Trimmomatic

Paired-end quality trimming was performed on all 12 samples using Trimmomatic v0.41 in PE (paired-end) mode, with Phred+33 quality encoding, using the parameters specified in the assignment:

- SLIDINGWINDOW:5:27 — scans the read with a 5-base window and trims from the point where the average quality within the window falls below Q27
- MINLEN:100 — discards any (trimmed) read shorter than 100 bp.
- AVGQUAL:27 — discards any read whose average quality across its full length is below Q27.

For each sample, Trimmomatic produced four output files: forward-paired, forward-unpaired, reverse-paired and reverse-unpaired reads, along with a per-sample log recording the trimming statistics. All 12 samples produced the expected 4 output files and 1 log file (48 trimmed FASTQ files and 12 logs in total).

Trimming Statistics Summary

Sample	Input Pairs	Both Surviving	Forward Only	Reverse Only	Dropped	Retention (%)
SRR14919235	135,499	133,906 (98.82%)	1,422 (1.05%)	136 (0.10%)	35 (0.03%)	98.82
SRR14919236	198,249	195,509 (98.62%)	2,503 (1.26%)	185 (0.09%)	52 (0.03%)	98.62
SRR14919237	103,416	102,290 (98.91%)	1,013 (0.98%)	90 (0.09%)	23 (0.02%)	98.91

SRR14919238	145,5 14	143,704 (98.76%)	1,624 (1.12%)	157 (0.11%)	29 (0.02%)	98.76
SRR14919239	132,9 14	131,234 (98.74%)	1,512 (1.14%)	129 (0.10%)	39 (0.03%)	98.74
SRR14919240	163,7 66	161,812 (98.81%)	1,760 (1.07%)	162 (0.10%)	32 (0.02%)	98.81
SRR14919241	189,8 69	188,123 (99.08%)	1,508 (0.79%)	208 (0.11%)	30 (0.02%)	99.08
SRR14919242	185,7 94	183,683 (98.86%)	1,874 (1.01%)	201 (0.11%)	36 (0.02%)	98.86
SRR14919243	157,5 97	156,078 (99.04%)	1,329 (0.84%)	161 (0.10%)	29 (0.02%)	99.04
SRR14919244	151,4 47	149,739 (98.87%)	1,519 (1.00%)	161 (0.11%)	28 (0.02%)	98.87
SRR14919245	151,0 24	149,268 (98.84%)	1,582 (1.05%)	136 (0.09%)	38 (0.03%)	98.84
SRR14919246	123,5 16	122,214 (98.95%)	1,128 (0.91%)	138 (0.11%)	36 (0.03%)	98.95

Inference from the Trimming Statistics

Across the full dataset, a total of 1,838,605 raw read pairs were processed, of which 1,817,560 pairs (98.86%) survived trimming with both mates intact, while only 407 pairs (0.022%) were dropped completely (neither mate surviving). This confirms the conclusion drawn in Question 1 that the raw data were of high overall quality, since only a very small fraction of reads failed the Q27 sliding-window/average-quality/length filters.

- Read retention is consistently high across all 12 samples, ranging from a minimum of 98.62% (SRR14919236) to a maximum of 99.08% (SRR14919241) of input pairs surviving as complete pairs — a narrow ~0.46 percentage-point spread, indicating uniform sequencing quality across the batch.
- Read loss (dropped pairs) is minimal throughout, ranging from 0.016% (SRR14919241) to 0.029% (SRR14919239). No sample shows an outlying or substantially higher drop rate, so no sample needed to be excluded from downstream analysis on quality grounds.
- Single-mate survivors (forward-only or reverse-only) are more common than fully-dropped pairs but still small (0.79–1.26% forward-only; 0.09–0.11% reverse-only). This asymmetry is expected because the reverse read of an Illumina 2× run typically has slightly lower average quality than the forward read, so the reverse mate is somewhat more likely to fail the AVGQUAL/MINLEN filters while its forward partner still passes.
- Effect of filtering parameters: the sliding-window trim (5 bp window, Q27 threshold) removes the low-quality 3' tail identified in the FastQC assessment without discarding whole reads; MINLEN:100 then removes any read trimmed down to an unusably short fragment, and AVGQUAL:27 provides a final safeguard against reads that are uniformly low-quality even if no single window triggered the sliding-window cut. Together these three filters are complementary: the sliding window handles localized quality drops, while MINLEN and AVGQUAL guard against short or globally poor reads.
- Downstream use: the ~98.6–99.1% of read pairs retained per sample as clean, quality-trimmed, properly-paired reads were carried forward into taxonomic classification with SPINGO (Question 3), ensuring that genus/species assignments are based on high-confidence base calls rather than low-quality 3'-end sequence.

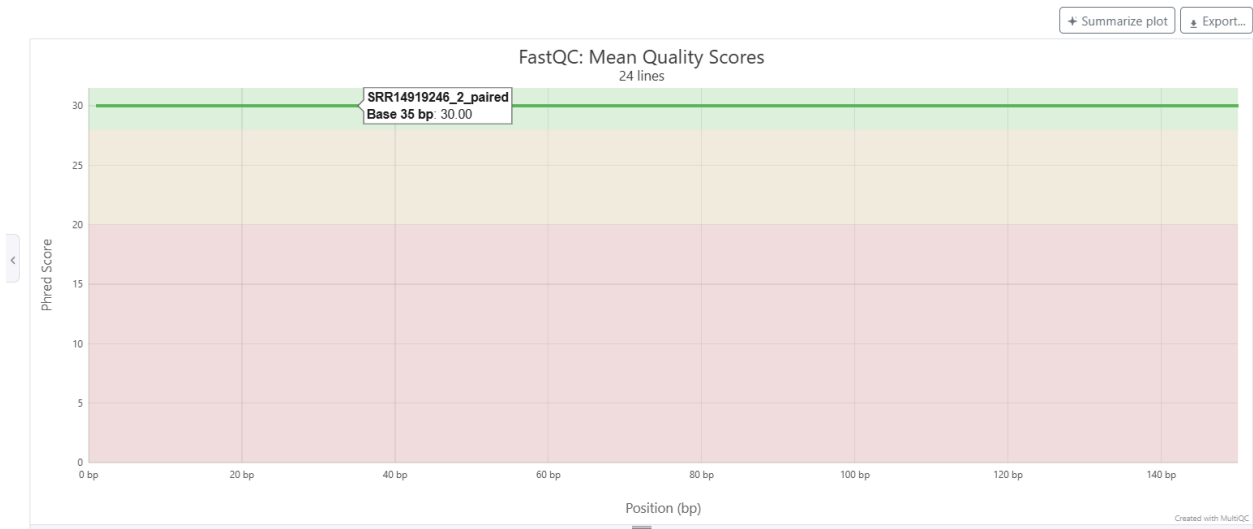
Post-Trimming Quality Check and Raw vs. Trimmed Comparison

FastQC was re-run on the 24 trimmed paired FASTQ files and the reports were again merged with MultiQC, producing a second, independent quality summary for comparison against the raw-read report from Question 1.

Quality Metric	Raw Reads	Trimmed Reads
Per-base sequence quality	Gradual quality decline toward the 3' end of most reads	3'-end low-quality regions were trimmed, resulting in an improved overall per-base quality profile; retained reads met the specified average-quality threshold of Q27
Sequence length distribution	Uniform, fixed raw read length	Variable length (sliding-window trimming), lower bound enforced at 100 bp by MINLEN
Per-sequence quality scores	High but with a low-quality tail from weaker reads	Distribution shifted right; only reads averaging ³ Q27 retained
Read pair counts	24 raw files, ~1.84 M pairs total	24 trimmed paired files, ~1.82 M pairs total (98.86% retained)
FastQC warnings	Occasional per-base-quality / adapter flags typical of raw data	Per-base quality flags reduced or resolved; residual, non-quality-related flags (e.g. sequence duplication, GC content) may persist since trimming does not address them

Table 2.2: Qualitative comparison of raw vs. trimmed read quality (from paired FastQC/MultiQC reports).

Trimming did not need to remove a large proportion of reads because the raw data were already high quality, but it did measurably improve the quality profile at the 3' end of the reads and enforced a consistent minimum length and average quality across the retained dataset. As expected, trimming does not resolve every FastQC flag (e.g., GC-content shape or sequence duplication level are intrinsic to the biological/library composition and are not affected by quality trimming) — only quality- and length-related flags are expected to improve. The resulting ~98.6–99.1% of properly paired, quality-trimmed reads per sample were judged suitable for downstream taxonomic classification with SPINGO.



FastQC: Mean Quality Scores

24 lines

SRR14919246_2_paired

Base 35 bp: 30.00

Phred Score

30

25

20

15

10

5

0

0 bp

20 bp

40 bp

60 bp

80 bp

100 bp

120 bp

140 bp

Position (bp)

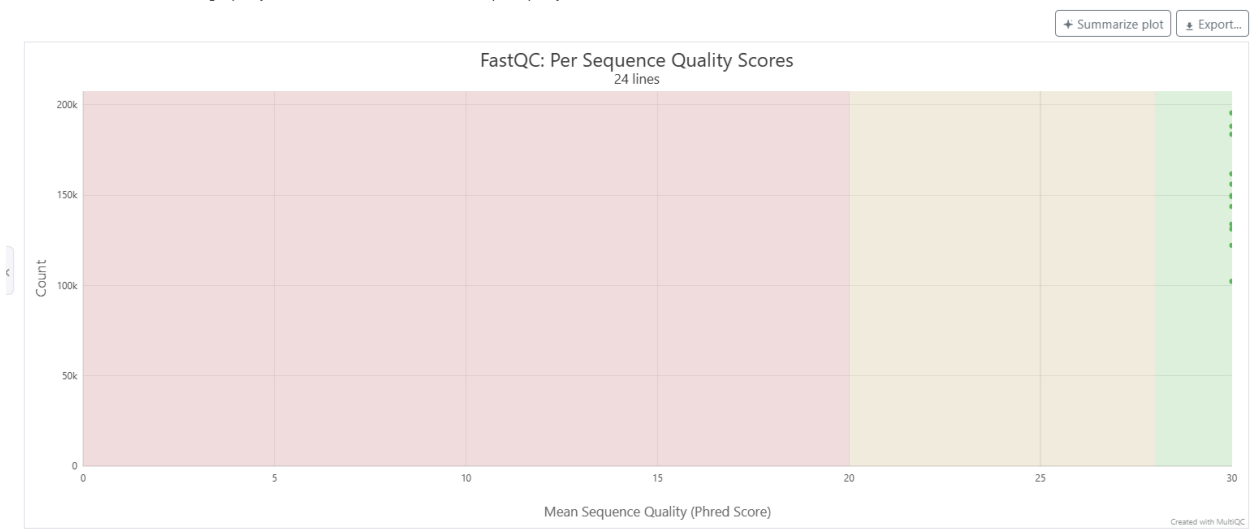
Created with MultiQC

Per Sequence Quality Scores

24

Help

The number of reads with average quality scores. Shows if a subset of reads has poor quality.



FastQC: Per Sequence Quality Scores

24 lines

Count

200k

150k

100k

50k

0

0

5

10

15

20

25

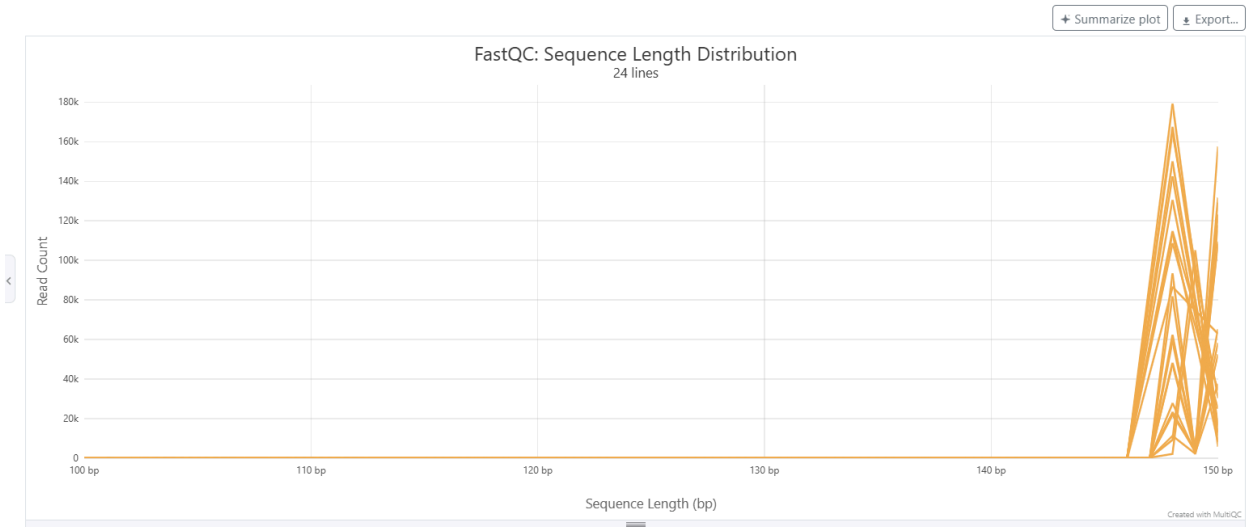
30

Mean Sequence Quality (Phred Score)

Created with MultiQC

Sequence Length Distribution 24

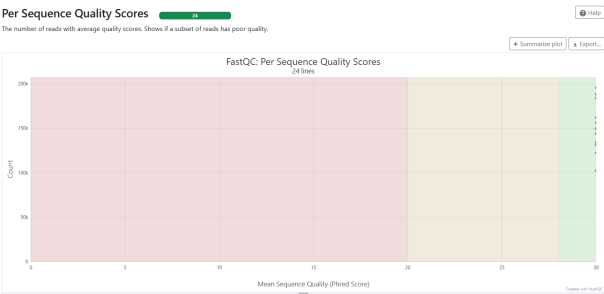
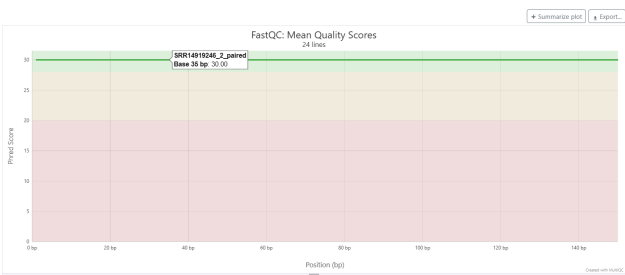
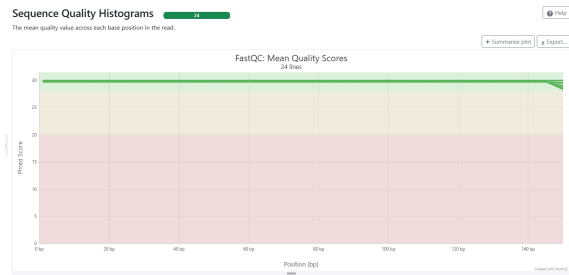
The distribution of fragment sizes (read lengths) found. See the [FastQC help](#)

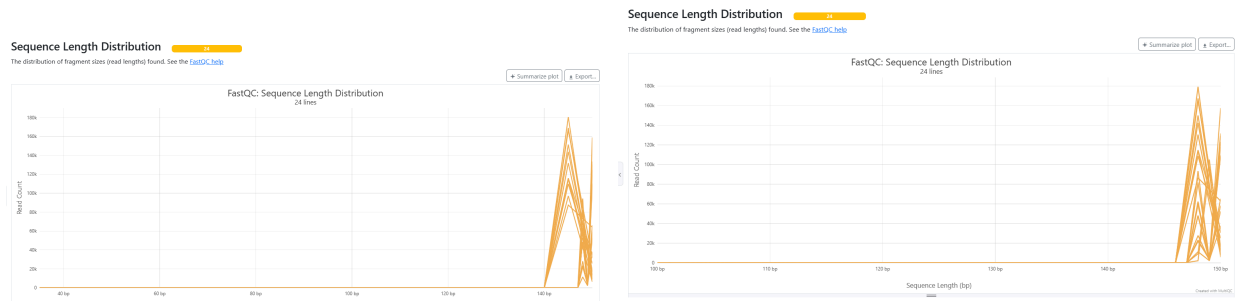


Comparison

Raw

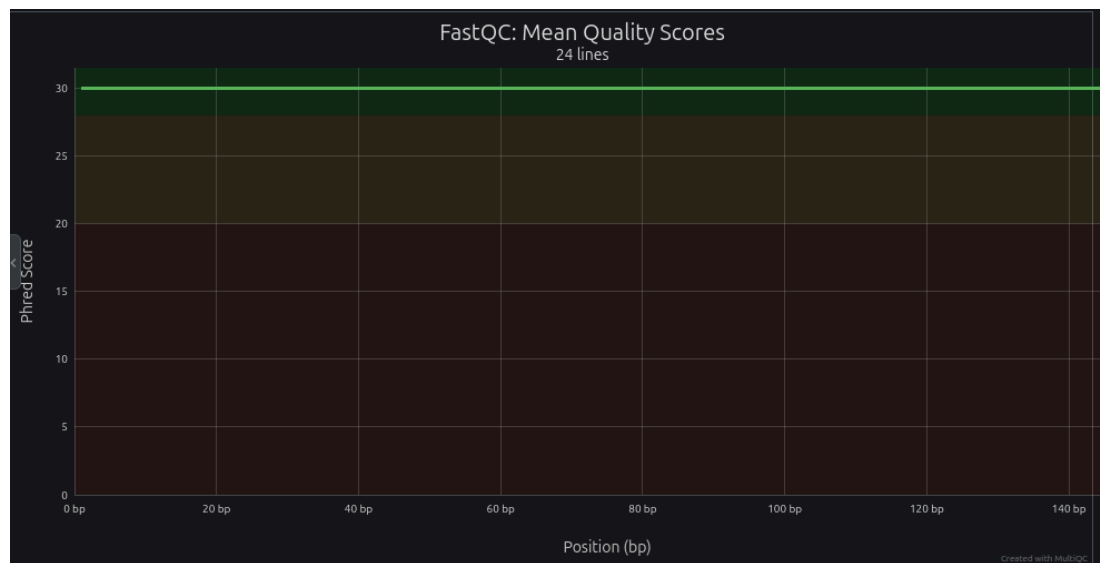
Trimmed

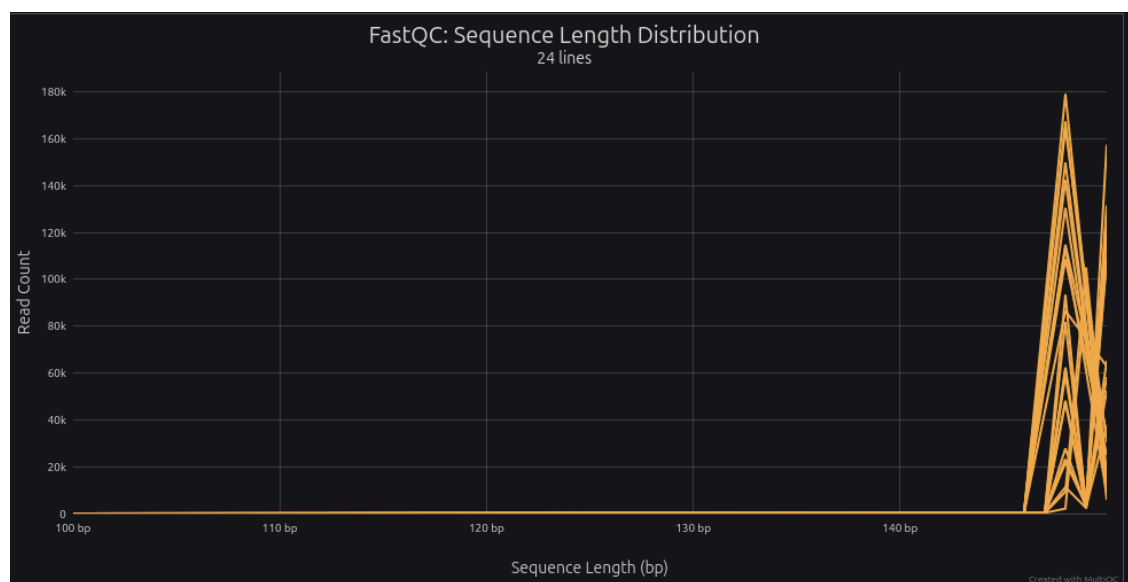
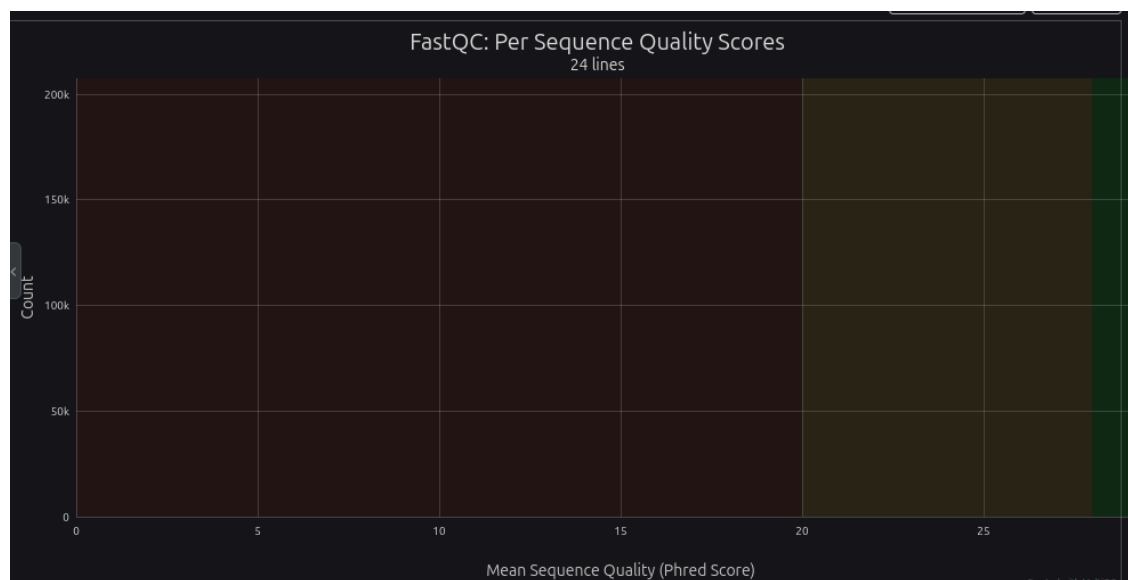




Cross-checking Trimming and QC Results with Galaxy Server

To cross-check the results on Galaxy Australia, the 12 raw paired SRA datasets were imported and organized into a paired dataset collection. Trimmomatic was executed on this collection using the sequential operations SLIDINGWINDOW:5:27, AVGQUAL:27, and MINLEN:100, generating 12 trimmed paired datasets alongside forward and reverse singletons. FastQC was then re-run on the 12 trimmed paired datasets to evaluate post-processing read quality improvements. Finally, MultiQC was executed, aggregating the Trimmomatic execution logs and FastQC zip outputs into a single consolidated report to verify the total read pair counts, survival rates, singletons, and dropped reads across all samples.





3. Using the sequences, classify the reads using SPINGO pipeline at both the genus and species levels. Compute the relative abundances of the species and normalize them using total-sum scaling normalization. Identify the top 20 species arranged based on their mean abundance and represent their abundances in sorted order using bar graphs. (5 marks)

Question 3: Taxonomic Classification with SPINGO

Method: Taxonomic classification was performed with SPINGO against the RDP 11.2 species-level reference database using the provided batch script (double_end_spingo.pl, 64 threads). For each sample, the quality-trimmed forward (_1) and reverse (_2) reads were concatenated and converted to FASTA, and SPINGO assigned every read simultaneously at the genus and species level, producing one tab-separated file per sample (SRRxxxxxxx.spingo.out.txt); column 5 holds the genus call and column 7 the species call, each with a bootstrap score.

From these per-read outputs, a species $\times$ sample count matrix (and a parallel genus $\times$ sample matrix) was built by tallying the reads assigned to each taxon in each sample. Reads returning AMBIGUOUS at a given rank, reads SPINGO could not confidently resolve to a single taxon, were not counted as a taxon at that rank; across the 12 samples, 75.5–84.9% of reads received a confident species-level assignment. Relative abundances were obtained by Total-Sum Scaling (TSS): each count was divided by its sample's total classified count, ensuring that all samples sum to 1 and making samples of differing sequencing depth directly comparable. Finally, the mean TSS abundance of each species across the 12 samples was computed, species were ranked by this mean, and the top 20 were plotted in sorted order (Figure 3.1), with the values in Table 3.1.

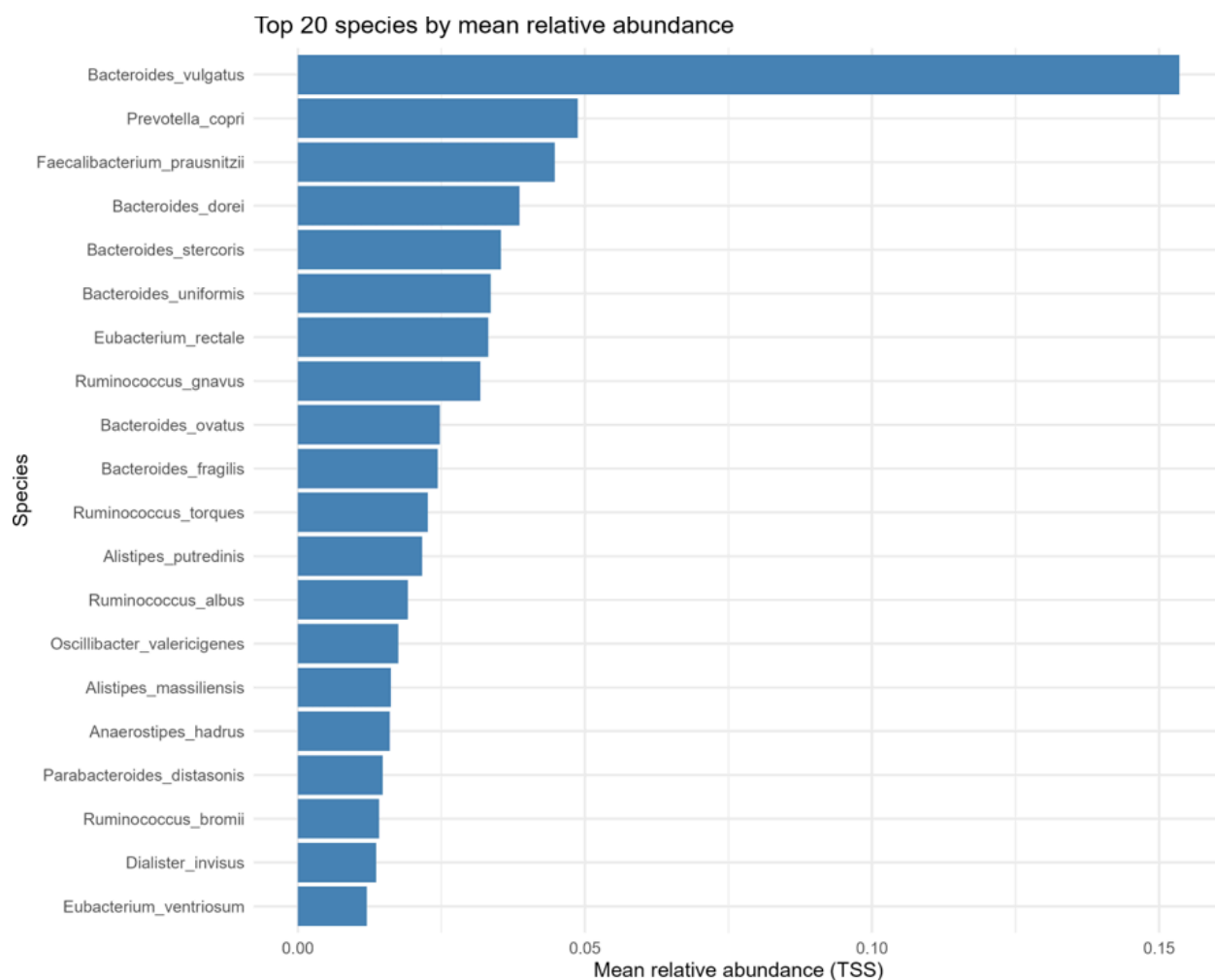


Figure 3.1: Top 20 species by mean TSS-normalised relative abundance across the 12 samples, shown in sorted (descending) order.

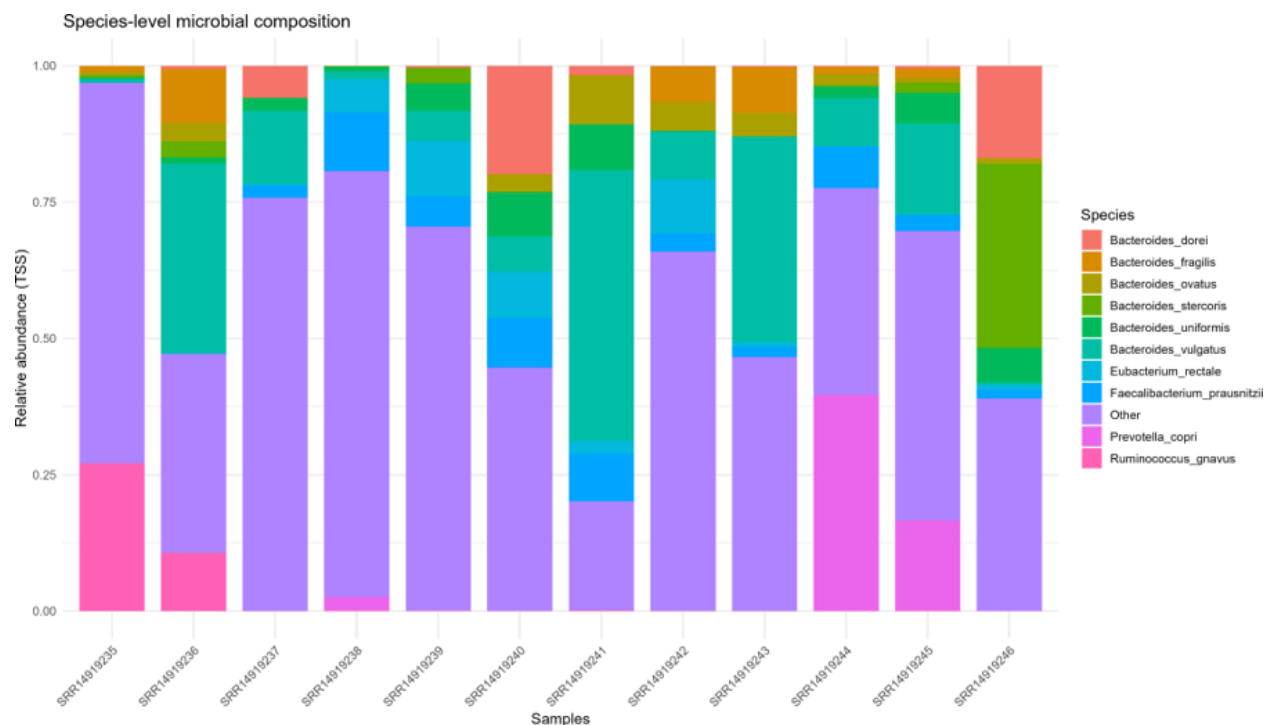
Interpretation

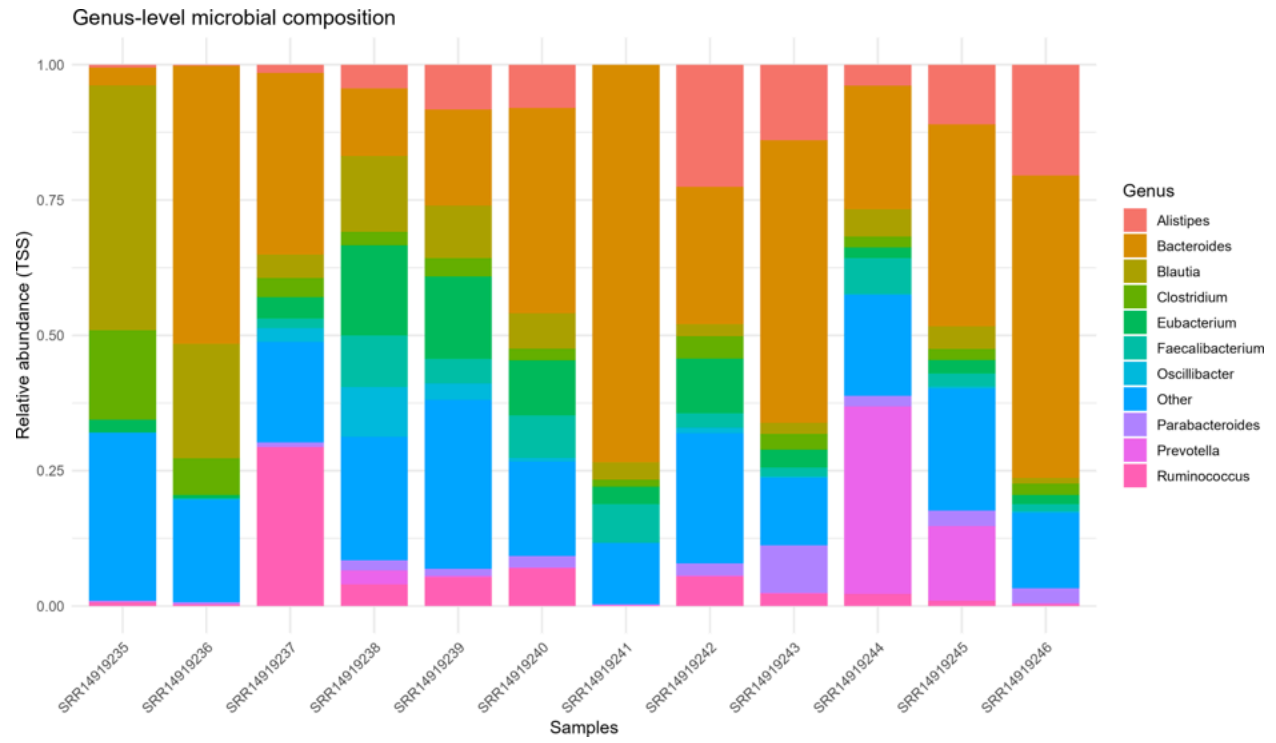
(Table 3.1: Mean TSS-normalised relative abundance of the top 20 species across the 12 samples.)

Species	Mean Relative Abundance (TSS)
<i>Bacteroides vulgatus</i>	0.1535
<i>Prevotella copri</i>	0.0488
<i>Faecalibacterium prausnitzii</i>	0.0448
<i>Bacteroides dorei</i>	0.0386
<i>Bacteroides stercoris</i>	0.0354
<i>Bacteroides uniformis</i>	0.0336
<i>Eubacterium rectale</i>	0.0331
<i>Ruminococcus gnavus</i>	0.0318
<i>Bacteroides ovatus</i>	0.0248
<i>Bacteroides fragilis</i>	0.0244
<i>Ruminococcus torques</i>	0.0227
<i>Alistipes putredinis</i>	0.0216
<i>Ruminococcus albus</i>	0.0191
<i>Oscillibacter valericigenes</i>	0.0175
<i>Alistipes massiliensis</i>	0.0162
<i>Anaerostipes hadrus</i>	0.0160
<i>Parabacteroides distasonis</i>	0.0148
<i>Ruminococcus bromii</i>	0.0141
<i>Dialister invisus</i>	0.0137
<i>Eubacterium ventriosum</i>	0.0120

Bacteroides_vulgatus was by far the most abundant species overall, with a mean relative abundance of 15.35% across the 12 samples, more than three times higher than the second-ranked species. *Prevotella_copri* (4.88%) and *Faecalibacterium_prausnitzii* (4.48%) were the next most abundant, followed by *Bacteroides_dorei* (3.86%) and *Bacteroides_stercoris* (3.54%). Notably, five of the top nine species by mean abundance belong to the genus *Bacteroides* (*vulgatus*, *dorei*, *stercoris*, *uniformis*, *ovatus*), consistent with *Bacteroides* being the dominant

genus in this cohort (see Question 4). *Faecalibacterium_prausnitzii* and *Eubacterium_rectale*, both well-known butyrate-producing commensals associated with a healthy gut, were also consistently among the most abundant species, while *Ruminococcus_gnavus* and *Ruminococcus_torques* (ranked 8th and 11th), taxa often linked with dysbiotic or inflammatory gut states, also feature among the most abundant species, indicating a mixed community profile across the cohort rather than a single uniform enterotype. Genus-level relative abundances were computed in parallel using the same TSS procedure, satisfying the requirement to classify at both the genus and species levels; the genus profiles are examined in detail in Question 4.





4. Compare the taxonomic profiles obtained at the genus and species levels across the 12 samples. Identify the major differences in microbial composition among the samples and discuss any taxa that are consistently abundant or uniquely detected in particular samples. Support your observations using appropriate plots or tables. (5 marks)

Comparison of Genus and Species Level Taxonomic Profile

The TSS-normalised genus and species level abundance matrices from Question 3 were compared across the 12 samples to identify:

- (i) Which taxa are consistently abundant or prevalent,
- (ii) Which taxa are uniquely or narrowly detected,
- (iii) Which samples share a similar overall profile, and
- (iv) Which samples differ most strongly in composition. Prevalence (number of samples with abundance > 0), mean/maximum relative abundance, and Bray–Curtis dissimilarity were used as the primary comparison metrics.

4.1 Dominant Taxon in Each Sample

Sample	Dominant Genus	Genus RA	Dominant Species	Species RA
SRR14919235	Blautia	0.452	Ruminococcus gnavus	0.271
SRR14919236	Bacteroides	0.514	Bacteroides vulgatus	0.348
SRR14919237	Bacteroides	0.335	Ruminococcus albus	0.178
SRR14919238	Eubacterium	0.166	Faecalibacterium prausnitzii	0.108
SRR14919239	Bacteroides	0.178	Eubacterium rectale	0.102
SRR14919240	Bacteroides	0.379	Bacteroides dorei	0.199
SRR14919241	Bacteroides	0.734	Bacteroides vulgatus	0.497
SRR14919242	Bacteroides	0.254	Alistipes massiliensis	0.133
SRR14919243	Bacteroides	0.522	Bacteroides vulgatus	0.377
SRR14919244	Prevotella	0.346	Prevotella copri	0.396
SRR14919245	Bacteroides	0.373	Bacteroides vulgatus	0.166
SRR14919246	Bacteroides	0.559	Bacteroides stercoris	0.337

Table 4.1: Single most abundant genus and species (relative abundance, RA, as a proportion of the sample) in each of the 12 samples.

- *Bacteroides* is the dominant genus in 10 of the 12 samples, ranging from a moderate 17.8% (**SRR14919239**) up to an extreme **73.4%** in SRR14919241, the most compositionally skewed sample in the dataset.
- The two exceptions are **SRR14919235**, dominated by *Blautia* (45.2%), and **SRR14919244**, dominated by *Prevotella* (34.6%).
- At the species level, *Bacteroides vulgatus* is the single most abundant species in 6 of the 12 samples (reaching 49.7% in SRR14919241 and 37.7% in SRR14919243), while other samples are led by *Ruminococcus gnavus* (SRR14919235), *Ruminococcus albus* (SRR14919237), *Faecalibacterium prausnitzii* (SRR14919238), *Eubacterium rectale* (SRR14919239), *Bacteroides dorei* (SRR14919240), *Alistipes massiliensis* (SRR14919242), *Prevotella copri* (SRR14919244) and *Bacteroides stercoris* (SRR14919246).

4.2 Prevalence and Consistently Abundant Taxa

Prevalence was computed as the number of the 12 samples in which each taxon's relative abundance was greater than zero. Of the 1,600 genera detected overall, **281 genera (17.6%)** were present in all 12 samples, while **367 genera (23%)** were detected in only a single sample. At the species level, of 4,413 species detected, **452 species (10.2%)** were present in all 12 samples, and **1,592 species (36%)** were detected in only one sample — the larger fraction of singleton detections at the species level reflects the finer taxonomic resolution and correspondingly greater sensitivity to low-depth, rare classifications.

Genus	Prevalence (/12)	Mean RA	Max RA	Sample of Max
Bacteroides	12	0.3528	0.7342	SRR14919241
Blautia	12	0.0990	0.4524	SRR14919235
Alistipes	12	0.0789	0.2254	SRR14919242
Eubacterium	12	0.0598	0.1663	SRR14919238
Ruminococcus	12	0.0483	0.2929	SRR14919237
Prevotella	12	0.0437	0.3462	SRR14919244
Clostridium	12	0.0411	0.1647	SRR14919235
Faecalibacterium	12	0.0381	0.0956	SRR14919238
Parabacteroides	12	0.0212	0.0878	SRR14919243
Oscillibacter	12	0.0149	0.0917	SRR14919238
Roseburia	12	0.0149	0.0493	SRR14919241
Anaerostipes	12	0.0147	0.0740	SRR14919235

Dialister	12	0.0131	0.0708	SRR14919236
Akkermansia	12	0.0100	0.0425	SRR14919239

Table 4.2: Top genera by mean relative abundance (RA), all detected in every sample (prevalence = 12/12).

Species	Prevalence (/12)	Mean RA	Max RA	Sample of Max
Bacteroides vulgatus	12	0.1535	0.4969	SRR14919241
Prevotella copri	11	0.0488	0.3957	SRR14919244
Faecalibacterium prausnitzii	12	0.0448	0.1077	SRR14919238
Bacteroides dorei	12	0.0386	0.1989	SRR14919240
Bacteroides stercoris	12	0.0354	0.3373	SRR14919246
Bacteroides uniformis	12	0.0336	0.0837	SRR14919241
Eubacterium rectale	12	0.0331	0.1020	SRR14919239
Ruminococcus gnavus	12	0.0318	0.2705	SRR14919235
Bacteroides ovatus	12	0.0248	0.0901	SRR14919241
Bacteroides fragilis	12	0.0244	0.0988	SRR14919236
Ruminococcus torques	12	0.0227	0.0993	SRR14919235
Alistipes putredinis	12	0.0216	0.0739	SRR14919246
Ruminococcus albus	12	0.0191	0.1775	SRR14919237
Oscillibacter valericigenes	12	0.0175	0.1033	SRR14919238

Table 4.3: Top species by mean relative abundance (RA); the top 13 species listed here are all detected in every sample (prevalence = 12/12).

Bacteroides (mean RA 35.3%), *Blautia* (9.9%), *Alistipes* (7.9%), *Eubacterium* (6.0%) and *Ruminococcus* (4.8%) are the five most abundant genera overall and, along with the remaining genera in Table 4.2, are consistently detected across the entire cohort — these represent the “core” genus-level microbiome of this dataset. At the species level, the 13 most abundant species (Table 4.3), led by *Bacteroides_vulgatus*, are likewise all detected in every one of the 12 samples, even though their relative abundance varies considerably between samples (e.g. *Bacteroides_vulgatus* ranges from a low background level up to 49.7% in SRR14919241). *Prevotella_copri*, although extremely abundant in specific samples (up to 39.6% in SRR14919244), is slightly less universal, being detected in 11 of the 12 samples.

Table 1. Ten highest-abundance genera uniquely detected in one sample

Genus	Relative Abundance	Sample Detected In
Fucsiella	0.00010	SRR14919237
Alloscardovia	0.00008	SRR14919246
Geoalkalibacter	0.00008	SRR14919239
Longilinea	0.00008	SRR14919239
Thermovirga	0.00008	SRR14919235
Stygiolobus	0.00007	SRR14919244
Halogeometricum	0.00007	SRR14919245
Aquifex	0.00007	SRR14919245
Iodobacter	0.00007	SRR14919240
Aciditerimonas	0.00007	SRR14919240

Table 2. Ten highest-abundance species uniquely detected in one sample

Species	Relative Abundance	
<i>Lactobacillus kitasonis</i>	0.00025	SRR14919239
<i>Eubacterium limosum</i>	0.00018	SRR14919235
<i>Streptococcus vestibularis</i>	0.00016	SRR14919240
<i>Campylobacter ureolyticus</i>	0.00015	SRR14919243
<i>Thalassomonas ganghwensis</i>	0.00014	SRR14919242
<i>Mesorhizobium loti</i>	0.00013	SRR14919241
<i>Fucsiella alikaci...</i>	0.00013	SRR14919237
<i>Prevotella dentasini</i>	0.00013	SRR14919238
<i>Actinomyces marimammalium</i>	0.00012	SRR14919244
<i>Nannocystis pusilla</i>	0.00011	SRR14919242

4.3 Top 15 and Top 20 Taxa Heatmaps Across Samples

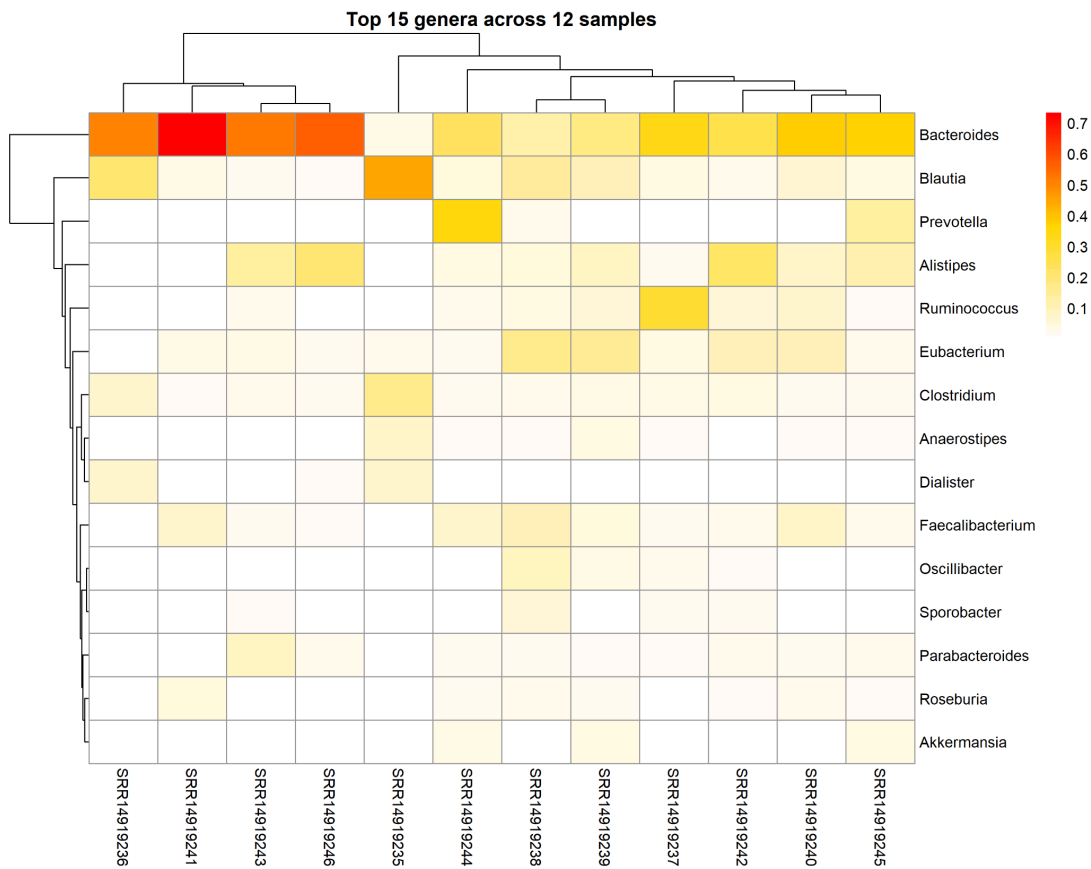


Figure 4.1: Relative abundance (TSS) of the top 15 genera across the 12 samples, with samples hierarchically clustered by similarity (column dendrogram).

The heatmap confirms that *Bacteroides* (top row) is the dominant genus with the highest abundance in most samples, but its abundance is clearly not uniform: it is deep red (highest abundance) in SRR14919241, SRR14919243, SRR14919236 and SRR14919246, but pale/absent in SRR14919235, where *Blautia* and *Clostridium* dominate instead. *Prevotella* stands out as almost exclusively confined to SRR14919244 (and, to a lesser extent, SRR14919245), illustrating a genus with a highly localized abundance pattern rather than a uniformly distributed one, even though it is technically prevalent (present at very low levels) in all samples.

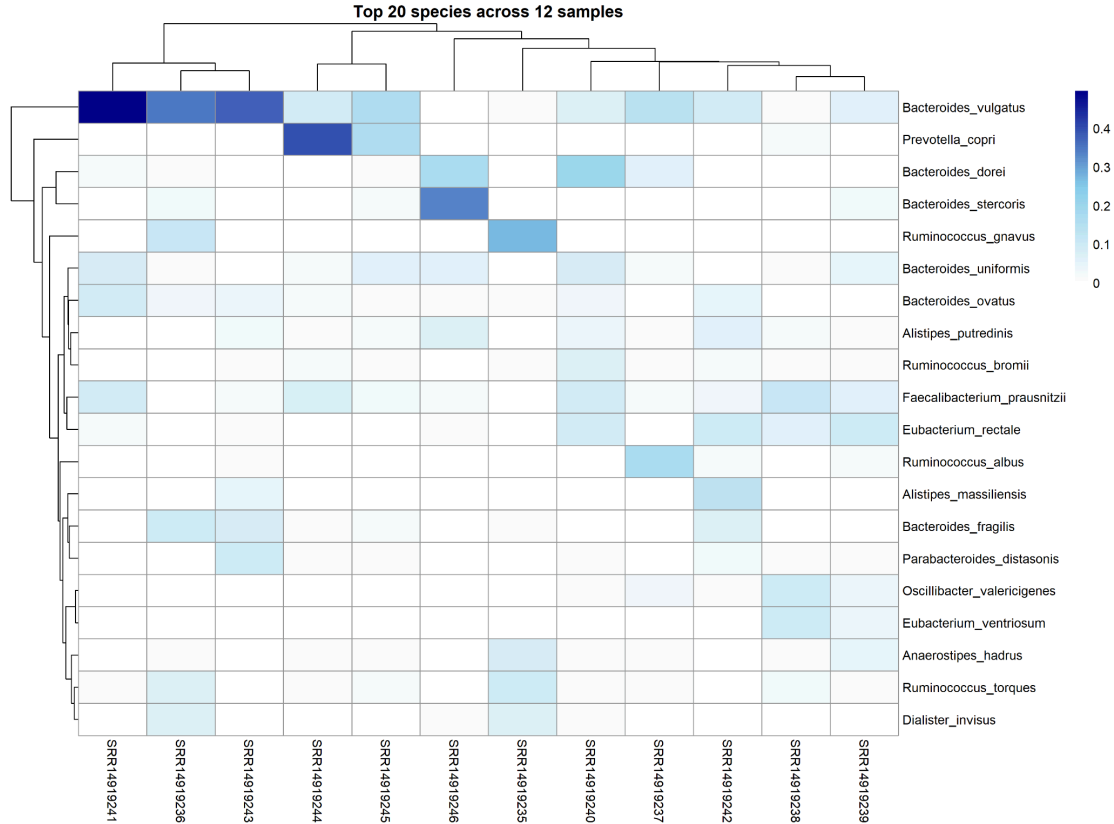


Figure 4.2: Relative abundance (TSS) of the top 20 species across the 12 samples, with samples hierarchically clustered by similarity (column dendrogram).

At the species level, the same pattern is visible with finer resolution: *Bacteroides_vulgatus* is the darkest, most abundant cell for SRR14919241, SRR14919236 and SRR14919243 (which cluster together on the dendrogram), while *Prevotella_copri* forms an isolated block of high abundance restricted to SRR14919244 (and moderately in SRR14919245). *Bacteroides_stercoris* is essentially unique to SRR14919246, and *Ruminococcus_gnavus* stands out specifically in SRR14919235. These sample-specific “hotspots” for otherwise widely-prevalent species explain much of the compositional dissimilarity quantified next.

4.4 Five Most Abundant Taxa per Sample

The five most abundant genera and species in every individual sample were also extracted (Tables 4.4 and 4.5 in the raw output files `top5_genera_by_sample.csv` and `top5_species_by_sample.csv`) to examine within-sample dominance structure. A condensed view is given below for two representative, contrasting samples.

Sample SRR14919241

Rank	Top Genera (RA)	Top Species (RA)
1	Bacteroides (0.734)	Bacteroides vulgatus (0.497)
2	Faecalibacterium (0.072)	Bacteroides ovatus (0.090)
3	Roseburia (0.049)	Faecalibacterium prausnitzii (0.087)
4	Eubacterium (0.032)	Bacteroides uniformis (0.084)
5	Blautia (0.032)	Roseburia faecis (0.040)

Sample SRR14919235

Rank	Top Genera (RA)	Top Species (RA)
1	Blautia (0.452)	Ruminococcus gnavus (0.271)
2	Clostridium (0.165)	Ruminococcus torques (0.099)
3	Anaerostipes (0.074)	Anaerostipes hadrus (0.080)
4	Parasutterella (0.068)	Parasutterella excrementihominis (0.078)
5	Dialister (0.067)	Dialister invisus (0.069)

SRR14919241 is an example of a low-diversity, single-taxon-dominated sample: its top genus (*Bacteroides*, 73.4%) and top species (*Bacteroides_vulgatus*, 49.7%) together already account for the bulk of the community, with the remaining top-5 taxa each below 9%. SRR14919235, by contrast, shows a more even, multi-taxon profile typical of a higher-diversity community, with its top five genera/species all in the 6–45% range and no single species exceeding 27%.

4.5 Sample Similarity: Bray–Curtis Dissimilarity

Bray–Curtis dissimilarity was calculated pairwise between all 12 samples, separately at the genus and species levels, using the TSS-normalized abundance matrices. A value of 0 indicates identical relative-abundance profiles, and a value approaching 1 indicates maximally dissimilar profiles.

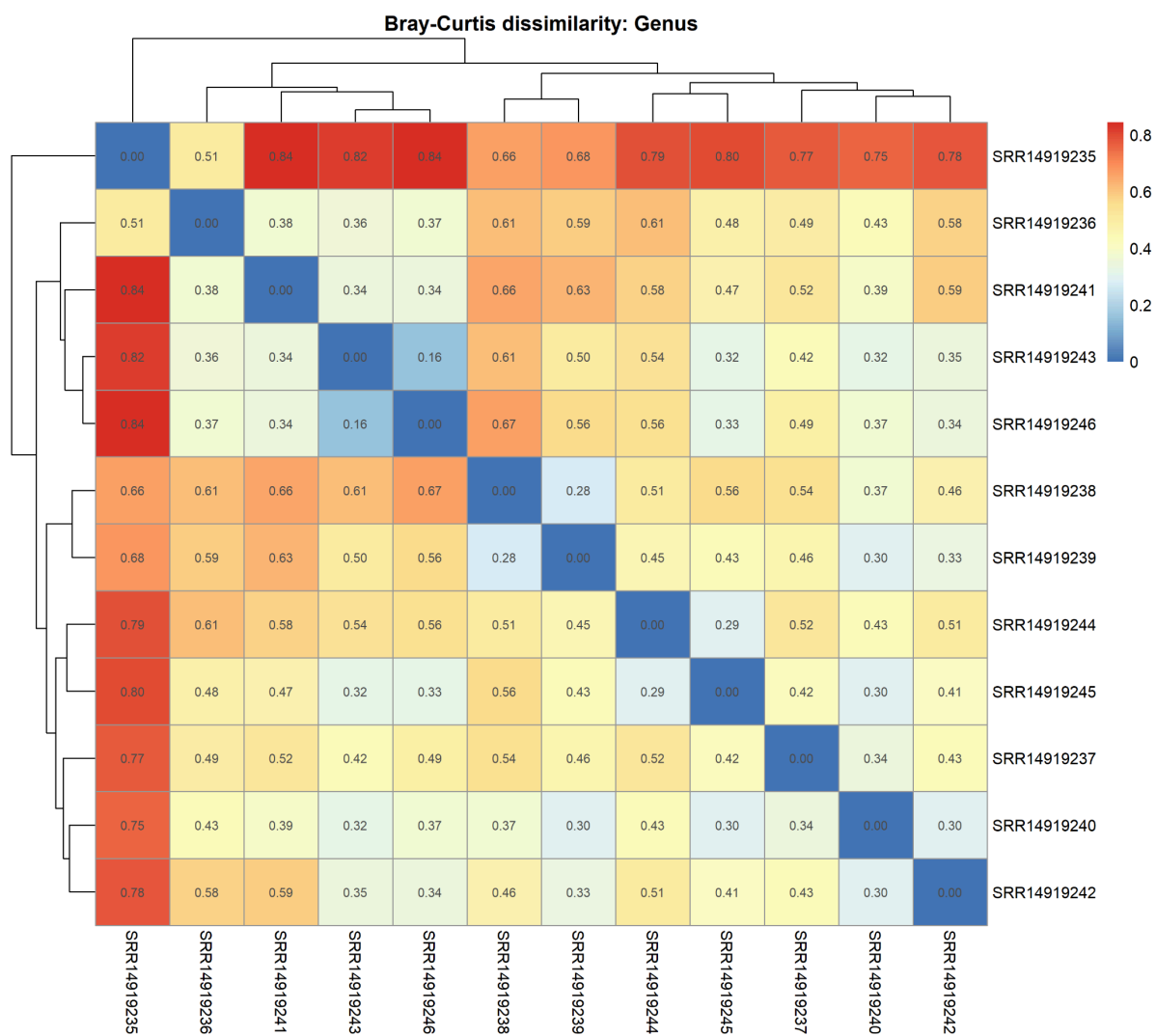


Figure 4.3: Genus-level Bray–Curtis dissimilarity matrix with hierarchical clustering (dendrogram) of the 12 samples.

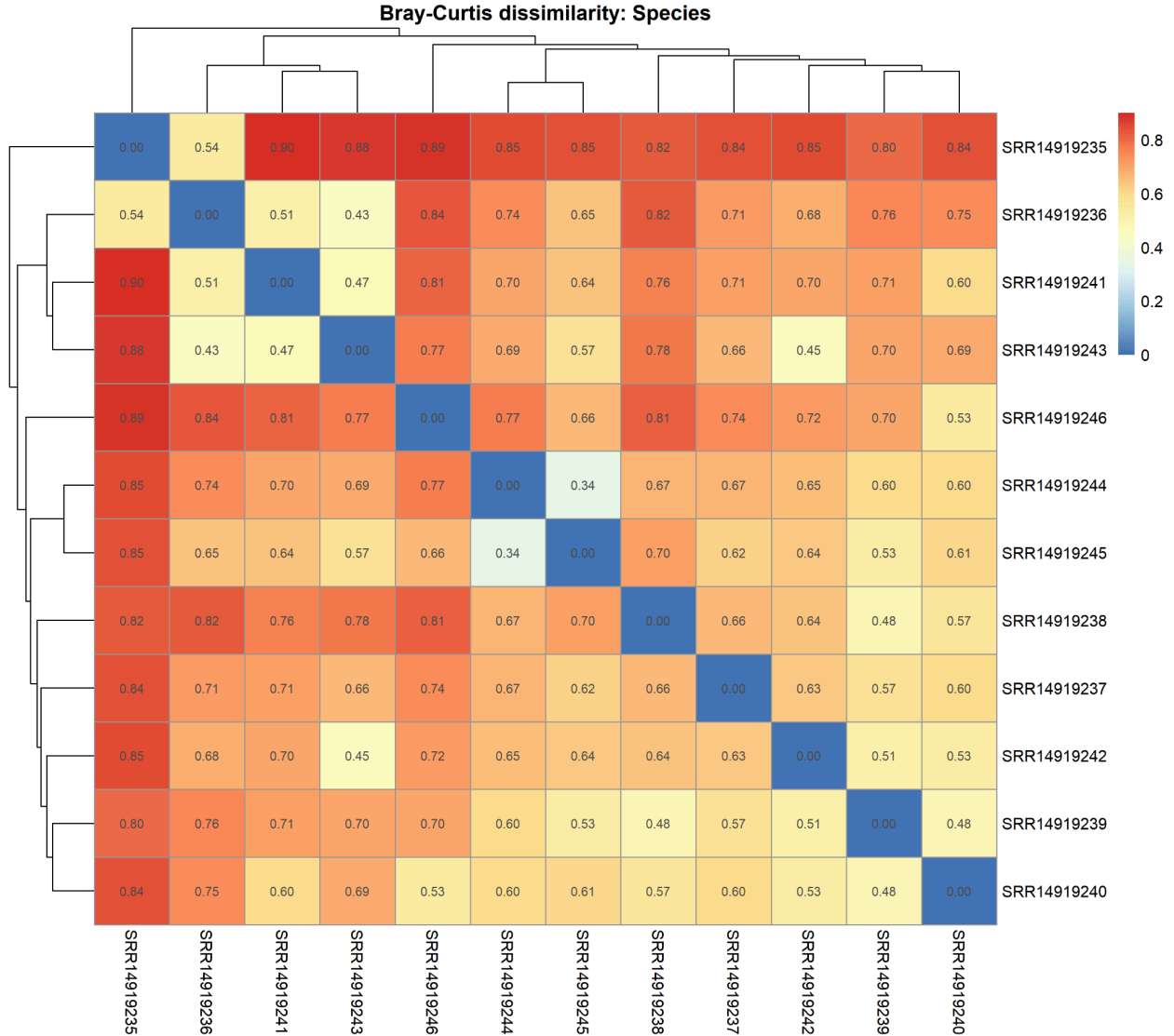


Figure 4.4: Species-level Bray–Curtis dissimilarity matrix with hierarchical clustering (dendrogram) of the 12 samples.

At the genus level, the clustering identifies two broad groups: a tight cluster of **SRR14919243** and **SRR14919246** (dissimilarity = **0.16**, the single lowest value in the whole matrix), reflecting their shared *Bacteroides*-dominated, *Alistipes*-secondary profile, and a second, looser cluster containing **SRR14919237**, **SRR14919240**, **SRR14919242**, **SRR14919244**, and **SRR14919245** (pairwise dissimilarities mostly 0.29–0.52). **SRR14919235** is the clear outlier, showing the highest dissimilarity to almost every other sample (0.66–0.84, and reaching 0.84 against both SRR14919241 and SRR14919246) — consistent with its unusual *Blautia*/*Clostridium*-dominated composition rather than the *Bacteroides*-dominated pattern seen elsewhere. The species-level matrix shows the same

overall structure but with uniformly higher dissimilarity values (mostly 0.45–0.90 instead of 0.16–0.84), which is expected because species-level profiles are more fine-grained and therefore more sensitive to differences between samples than the coarser genus-level view; SRR14919235 remains the most dissimilar sample overall (species-level dissimilarity of 0.80–0.90 to most other samples), and SRR14919243/SRR14919246 again form one of the more similar pairs (species-level dissimilarity = 0.77).

4.6 Overall Comparison and Discussion

Combining the results from Sections 4.1–4.5, the 12 samples separate into a clear compositional gradient rather than sharp discrete groups: at one extreme, samples such as SRR14919241, SRR14919243 and SRR14919246 are strongly dominated by a single *Bacteroides* species (*vulgatus* or *stercoris*) at 34–74% relative abundance, forming a tight, low-dissimilarity cluster; at the other extreme, SRR14919235 (*Blautia*/*Clostridium*/*Ruminococcus_gnavus*-led) and SRR14919244 (*Prevotella_copri*-led) stand out as compositionally distinct outliers with high pairwise dissimilarity to the rest of the cohort. Genus-level comparisons smooth over some of this variation (because closely related *Bacteroides* species are pooled into a single genus), which is why species-level dissimilarities are consistently higher than the corresponding genus-level values for the same sample pairs. A small set of taxa — *Bacteroides*, *Blautia*, *Alistipes*, *Eubacterium*, *Ruminococcus* at the genus level, and *Bacteroides_vulgatus*, *Prevotella_copri*, *Faecalibacterium_prausnitzii* at the species level — are consistently abundant across essentially the entire cohort and form the shared “core” microbiome, whereas taxa such as *Prevotella_copri*/*Prevotella* and *Bacteroides_stercoris* illustrate taxa that, despite being broadly prevalent at low levels, reach dominant abundance in only one or two specific samples. The very large number of singleton (prevalence = 1) genera and species reflects normal low-abundance/rare-taxon detection in amplicon sequencing rather than a biologically meaningful difference between samples.

4.7 Conclusion

Comparing the genus- and species-level SPINGO taxonomic profiles across the 12 samples showed that *Bacteroides* and its constituent species, especially *Bacteroides_vulgatus*, dominate the majority of samples and account for the largest share of mean relative abundance dataset-wide, while a core set of genera and species (*Blautia*, *Alistipes*, *Eubacterium*, *Ruminococcus*, *Faecalibacterium_prausnitzii*, *Eubacterium_rectale*, among others) are consistently detected in every sample and form a shared core community. At the same time, distinct sample-specific dominance patterns — particularly the *Blautia*-led

profile of SRR14919235 and the *Prevotella_copri*-led profile of SRR14919244 — drive the majority of the compositional dissimilarity captured by the Bray–Curtis analysis, and the close clustering of SRR14919243/SRR14919246 illustrates that some samples nonetheless share highly similar overall community structure.

Group Contribution Summary

Task	Description	Group Member Responsible
Q1	Raw-read quality assessment with FastQC / MultiQC	ANIKET KHANDELWAL
Q2	Quality trimming with Trimmomatic and post-trim QC	Ponnamperuma Arachchige Ruchila Ranjula Wijerathne
Q3	SPINGO taxonomic classification, TSS normalisation, top-20 species analysis	AKSHAT KUMAR
Q4	Genus/species comparison, prevalence analysis, Bray–Curtis dissimilarity	K Sriram
Report	Compilation of README, results and final PDF report	Sunandini Chowdhury